The Arithmetic Uniqueness of the Standard Model: Asymptotic Sparsity, Galois Structure, and the Positive-Root Theorem

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[1]Independent Research

2026

Preprint.

Archival version (Zenodo): [DOI pending publication]

Abstract

The Universal Generative Principle (UGP) addresses the Standard Model parameter spectrum, classifying each sector by epistemic status [1], from a uniquely selected integer seed at ridge level n=10 via a deterministic arithmetic cascade. We investigate whether UGP is itself a shadow of a deeper mathematical structure.

Four main results. (1) The **Asymptotic Sparsity Theorem** proves the joint arithmetic-admissibility and physical-viability constraint has exactly one solution across all ridge levels: (n=10, b_1=73). The arithmetic component is Lean-certified (unconditional); the physical-viability component uses the CODATA-derived instantiation factor (A/D). (2) The **Positive Root Theorem** identifies an exact arithmetic correspondence between the SM bare gauge-coupling numerators and the positive-root counts $|\Phi^+(G)|$ of the respective gauge groups; the **Chirality Theorem** shows the squareness/non-squareness of these numerators is the Lean-certified arithmetic signature of the vector-like/chiral distinction. (3) **Galois structure**: the UGP algebraic constants lie in the unique minimal cyclotomic field $\mathbb{Q}(\zeta_{120})$ whose conductor is the lcm of the Coxeter numbers of the SM gauge algebras; algebras with $h \nmid 120$ (notably E_7, $h = 18$) are *algebraically impossible* to contain in $\mathbb{Q}(\zeta_{120})$ — a structural Tower Law theorem, Lean-certified zero-sorry (Corollary 7.5). (4) The **WZW route is falsified**: the Wess-Zumino-Witten construction cannot reproduce the UGP bare coupling rationals, closing a natural alternative-parent hypothesis.

Epistemic status: all main arithmetic theorems are $\mathrm{A}_{\mathrm{Lean}}$ (Lean-certified, zero `sorry`); physics-bridge interpretations and CODATA-conditioned results are A/D. A front-matter claim-status table lists every result's status explicitly.

All computations are reproducible via the open-source scripts at `https://github.com/novaspivack/ugp-physics` (`papers/24_deeper_theory/`; `run_all.py` reproduces all results in < 1 s).

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Contents

1 Introduction

The Universal Generative Principle (UGP) [1] is a deterministic number-theoretic framework that addresses the Standard Model parameter spectrum, classifying each sector by epistemic status, from three axioms — locality, symmetry, and compression (MDL) — via a uniquely selected integer seed at ridge level n=10. Its Lean 4 formalization [2,3] (see companion formalization paper [2], zero `sorry`, zero custom axioms beyond Mathlib) machine-checks the core derivation chain, including the Residual Seed Uniqueness Classification (RSUC) theorem that isolates the lepton seed $(1, 73, 823)$ as the unique MDL-minimal survivor of the two-stage arithmetic sieve.

The framework has produced a series of certified structural results: the blind α_s prediction at $+0.24\sigma$ of PDG 2024 [1], the neutrino mass-squared ratio $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$ matching NuFIT 6.0 [4] to 0.16σ (0.52%) [5], the Koide closed form predicting m_τ from (m_e, m_μ) at 61 ppm [6], and the Two-Layer PSC Theorem establishing the Standard Model as PSC-optimal among all 4D gauge theories [7].

The question of why the Standard Model has its specific gauge group and representation content has been approached from several mathematical directions that are complementary to the present work: via noncommutative spectral geometry and arithmetic motives [8,9], via exceptional Jordan algebras and octonions [10, 11], and via division algebras [12]. The present paper addresses a related selection question through arithmetic uniqueness: the UGP seed is the unique solution of a Diophantine sieve, machine-certified for all $n \in \mathbb{N}$.

A natural question arises: is UGP itself a shadow of a deeper mathematical structure? The recurring patterns — the same integers ($N_c = 3$, $\delta = 7$, $29 = 4N_c^2 - \delta$) appearing in algebraically independent sectors, the golden ratio φ throughout the kernel, the cyclotomic angles $\pi/6$, $\pi/10$, $\pi/12$ — suggest that UGP might be a coordinate representation of a more fundamental object.

This paper reports the results of a systematic investigation of this question. We tested the hypothesis that UGP is the shadow of a specific WZW modular invariant, found it falsified, and identified the correct characterization: UGP is the unique rational point on the intersection of two independent constraint systems.

1.1 Summary of Main Results

Theorem 1.1 (Asymptotic Sparsity, informal)**.** *The joint constraint of arithmetic admissibility (Stage 1 sieve) and physical viability (Stage 2 δ-match) has exactly one solution across all $n \in \mathbb{N}$: (n=10, b_1=73, seed $= (1, 73, 823)$).*

Machine-checked: `asymptotic_sparsity_universal`*, module* `Phase4.AsymptoticSparsity`*,* `ugp-lean` *[3], zero* `sorry`*.*

Theorem 1.2 (Positive Root Theorem)**.** *The number of* $\mathrm{SU}(N)_1$ *WZW factors in the numerator of the bare gauge coupling* g_G^2 *equals* $|\Phi^+(G)|$*, the number of positive roots of the gauge group* G*.*

Machine-checked: `positive_root_theorem`*, module* `Phase4.PositiveRootTheorem`*,* `ugp-lean` *[3], zero* `sorry`*.*

Theorem 1.3 (Chirality Theorem)**.** *The* $\mathrm{SU}(3)$ *coupling numerator* $(13 \cdot 17 \cdot 29)^2$ *is a perfect square while the* $\mathrm{SU}(2)$ *numerator* $17 \cdot 137$ *is not. This is the arithmetic signature of the* $V-A$ *structure:* $\mathrm{SU}(3)$ *is vector-like (both chirality sectors contribute, squaring the factor);* $\mathrm{SU}(2)$ *is chiral (one sector, no squaring).*

Machine-checked: `chirality_arithmetic`*,* `BraidAtlas.ChiralitySquaring`*,* `ugp-lean` *[3], zero* `sorry`*.*

Theorem 1.4 (Galois Stability)**.** *All UGP algebraic constants live in* $\mathbb{Q}(\zeta_{120})$*. The UGP layers (kernel, TT/*A_2*, Koide) are Galois-stable subsets. Galois orbit sizes are uniformly equal to* strand_count $= (N_c^2 - 1)/4 = 2$*.* $120 = \mathrm{lcm}(20, 24)$ *is the minimal cyclotomic conductor for all UGP constants.*

Machine-checked: `galois_layer_stability`*,* `q_zeta_120_is_minimal_conductor`*, modules* `GaloisStructure.CyclotomicLayers`*,* `GaloisStructure.MinimalCyclotomic`*,* `ugp-lean` *[3], zero* `sorry`*.*

Theorem 1.5 (Fibonacci-Mersenne Structure)**.** $n_{\mathrm{ridge}} = 10 = 2F(5)$ *is the exponent of the second Mersenne rung. The rung exponents* $\{4, 10, 16\} = \{2F(3), 2F(5), 2F(6)\}$ *form an arithmetic sequence with step* $2F(4) = 2N_c = 6$*.*

Machine-checked: `mersenne_ladder_structure`*,* `n_ridge_structural_position`*, module* `GaloisStructure.MinimalCyclotomic`*,* `ugp-lean` *[3], zero* `sorry`*.*

Theorem 1.6 (VV Mechanism)**.** *The VV log-linear down-quark mass relation arises from composing the SU(5)/SO(10) Yukawa power law* $Y_d \propto Y_u^{13/9} \cdot Y_\ell^{-7/6}$ *with the UCL logarithmic mass map.*

Machine-checked: `vv_mechanism_algebraic`*, module* `MassRelations.VVMechanism`*,* `ugp-lean` *[3], zero* `sorry`*.*

Theorem 1.7 (Origin of 137)**.** $137 = 2^0 + 2^{N_c} + 2^\delta$ *is the bit-set encoding of the UGP atomic parameters* $\{0, N_c, \delta\} = \{0, 3, 7\}$*, fully determined by* $N_c = 3$*.*

Machine-checked: `prime_137_structural_origin`*,* `GaloisStructure.MinimalCyclotomic`*,* `ugp-lean` *[3], zero* `sorry`*.*

Theorem 1.8 (Galois-Protection Cancellation Lemma)**.** *If a one-loop contribution to a Galois-stable UGP quantity appears in a* $T/T^\dagger$*-paired form* $L + (-L)$ *where all transcendentals lie outside* $\mathbb{Q}(\zeta_{120})$ *(Baker's theorem), then the net one-loop correction vanishes. Machine-checked:* `galois_protection_master_theorem`*,* `Phase4.GaloisProtection`*,* `ugp-lean` *[3], zero* `sorry`*. Status: This theorem proves the* $\boldsymbol{A}_{\mathrm{Lean}}$ *algebraic cancellation lemma. The claim that the physical one-loop QED effective action for* α_{EM} *realizes precisely this* $T/T^\dagger$*-paired form is an* ***A/D*** *physics bridge; a direct effective-action derivation establishing this realization remains open.*

Theorem 1.9 (Two-Loop Color Coefficient ($\mathrm{A}_{\mathrm{Lean}}$))**.** *The* $\mathrm{SU}(N_c)$ *adjoint color coefficient* $(N_c^2 - 1)/N_c^2$*, derived from* $N_c = 3$ *alone via the quadratic Casimir* $C_2(\mathrm{fund}) \times 2/N_c$*, equals* $8/9$*. Machine-checked:* `two_loop_coefficient_eq_8_over_9`*,* `Phase4.TwoLoopCoefficient`*,* `ugp-lean` *[3], zero* `sorry`*. The coefficient* $8/9$ *is certified from* $N_c = 3$ *alone; no reference to the measured residual enters the Lean proof.*

A/D physical identification. The proposed two-loop residual bridge is

$$R_{\mathrm{pred}} = (8/9) \times \alpha_{\mathrm{EM}}^2/(2\pi^2) \approx 2.40 \text{ ppm},$$

which matches the CODATA-derived residual 2.39 ppm at 0.33%. The 8/9 coefficient is $\mathrm{A}_{\mathrm{Lean}}$; the full perturbative identification of the physical effective two-loop term with exactly this formula remains **A/D** pending a direct effective-action derivation.

Falsification (T4). The modular invariant $Z(\tau)$ of $\mathrm{SU}(2)_8 \otimes \mathrm{SU}(3)_3 \otimes \mathrm{SU}(2)_{10}$ does not encode the bare gauge couplings as Fourier coefficients. The WZW connections are algebraic shadows of $\mathbb{Q}(\zeta_{240})$, not a parent theory.

1.2 Organization

Throughout this paper we use the five-status claim taxonomy of the companion paper [1]:

- $\mathbf{A}_{\text{Lean}}$: Lean-certified arithmetic or structural identity (zero `sorry`).
- $\mathbf{A}_{\text{MDL}}$: MDL-uniqueness certificate within a declared expression class; not Lean-derived from invariants.
- [**A/D**]: A Lean-certified arithmetic statement plus an explicit physics bridge, external scale, or interpretation not itself certified in Lean.
- [**B**]: Computationally supported / empirically verified (PSLQ, numerical scan).
- [**D**]: Speculative or frontier.

Plain **A** in theorem labels means A_{Lean}. The main theorems in this paper (Asymptotic Sparsity arithmetic part, Positive Root, Chirality squareness, Galois Stability, lcm identity, Lean arithmetic lemmas) are A_{Lean}. Stage-2 joint-uniqueness, Galois-protection physics bridge, two-loop residual physical identification, and E8 positive cases are A/D or B as labeled.

Table 1: Claim-status map for this companion paper.

Result	Status	Notes
Asymptotic Sparsity (arithmetic)	$\mathrm{A}_{\mathrm{Lean}}$	$\forall n$: mirror-dual survivor with $b_1 = 73$ forces $n = 10$.
Stage-2 joint uniqueness	A/D	CODATA-conditioned; arithmetic sieve is exact, δ_{target} is external.
Structural prediction $C_{\mathrm{alg}}/73$	A/D	$b_1 = 73$ and C_{alg} are structural; CODATA comparison is empirical validation.
Positive Root Theorem	$\mathrm{A}_{\mathrm{Lean}}$	Exact arithmetic correspondence over SM gauge groups; three data points.
Chirality squareness/non-squareness	$\mathrm{A}_{\mathrm{Lean}}$	Lean arithmetic. V−A interpretation is Braid-Atlas bridge (A/D).
Galois Stability / lcm	$\mathrm{A}_{\mathrm{Lean}}$	Cyclotomic containment and lcm identity Lean-certified.
Positive Coxeter embeddings (G2, F4, E6, E8)	$\mathrm{A}_{\mathrm{Lean}}$/B	$\mathbb{Q}$-algebra embedding $\mathrm{CyclotomicField}(2h, \mathbb{Q}) \hookrightarrow \mathrm{CyclotomicField}(120, \mathbb{Q})$ Lean-certified (`cyclotomic_field_embedding`, `CyclotomicContainment`); mass formulas from integrable-QFT literature.
E7 Tower Law exclusion ($h = 18 \nmid 120$)	$\mathrm{A}_{\mathrm{Lean}}$/B	Algebraically impossible by Tower Law — Lean-certified zero sorry (`e7_tower_law_complete`) + PSLQ.
Galois protection (algebraic)	$\mathrm{A}_{\mathrm{Lean}}$	Abstract $L + (-L)$ cancellation lemma.
Galois protection (physical)	A/D	Physical one-loop realization of $T/T^\dagger$ pairing is bridge-dependent. Galois-Selection Condition (Prop. 9.1) gives the defensible constraint-based form; full QFT-level proof is **O2**.
Two-loop (8/9) coefficient	$\mathrm{A}_{\mathrm{Lean}}$	$(N_c^2 - 1)/N_c^2$ from $N_c = 3$, Lean-certified.
Two-loop residual (physical)	A/D	Full field-theoretic identification of the effective two-loop term is bridge-dependent.
RCC / classical families	$\mathrm{A}_{\mathrm{Lean}}$	Lean-certified Lie-algebraic argument. See App. D.
RCC / exceptional sector	A/D + B	Lean (layer definitions) + computational scan.
WZW falsification	A + B	Arithmetic mismatch (Lean) + numerical verification.

Section 2 reviews the UGP framework and the seed-selection problem. Section 3 formulates the joint constraint as a Diophantine system. Section 4 proves the Asymptotic Sparsity Theorem. Section 5 establishes the Positive Root Theorem. Section 6 resolves the chirality question. Section 7 proves Galois stability. Section 8 summarizes the WZW investigation and its falsification. Section 9 presents the deeper law, the Fibonacci-Mersenne structure, the VV mechanism, the minimal cyclotomic substrate, and the arithmetic origin of 137. Section 9 also collects three additional structural results that sit naturally inside the deeper-theory programme: Section 9.5 derives the SM winding numbers $\{N_c{-}1, -1, 0, -N_c\}$ from the colour rank alone (`BraidAtlas.ChargeTheorem`); Section 9.6 establishes the Residual Classification (RCC) as a theorem over all compact simple Lie groups (`PSC.RCCInfiniteFamilies`, plus an extended TE2.2 certificate); and Section 9.7 audits the boundary of UGP-internal derivability, showing that anomaly cancellation is the unique constraint that forces $N_c = 3$ without circularity within the present framework. Section 9.10 establishes the full Precision Derivation Programme: the one-loop correction to C_{alg} vanishes by Galois-protection (Lean: `galois_protection_master_theorem`) and the surviving two-loop residual is

$R_{\text{real}} = (8/9) \times \alpha_{\text{EM}}^2/(2\pi^2)$ with $(8/9)$ Lean-certified (`Phase4.TwoLoopCoefficient`, zero `sorry`). Section 10 concludes.

2 Background: UGP and the Seed-Selection Problem

2.1 The UGP Framework

The UGP is founded on three axioms:

A1 (Locality): All fundamental interactions are local in the substrate's causal graph.

A2 (Symmetry): The fundamental laws are invariant under a set of discrete transformations.

A3 (Compression/MDL): The universe evolves according to rules that minimize total description length.

Its dynamical realization, the Generative Triple Evolution (GTE), operates on integer triples $(a, b, c;\ g) \in \mathbb{Z}^3 \times \{1, 2, 3\}$. The components encode: a = interaction complexity (M"obius phase); b = ladder index (spacetime volume, related to mass); c = Mersenne-ladder capacity (dominant frequency, chirality via sign); g = generation.

The GTE cascade from the lepton seed $(1, 73, 823)$ generates all SM fermion triples:

$$(1, 73, 823;\ 1) \xrightarrow{\text{odd}} (9, 42, 1023;\ 2) \xrightarrow{\text{even}} (5, 275, 65535;\ 3),$$

with quark seeds derived by permutation. The Universal Calibration Law (UCL) maps each triple to a physical mass via a nine-component kernel whose algebraic structure is fixed by the Quarter-Lock identity:

$$k_M \ = \ k_{\text{gen2}} + \tfrac{1}{4}\, k_{L^2}, \tag{1}$$

where $k_{L^2} = 7/512$, $k_{\text{gen2}} = -\varphi/2$, and $\varphi = (1 + \sqrt{5})/2$. This identity is machine-checked in `ugp-lean` as `quarterLockLaw` (zero `sorry`).

2.2 The Seed-Selection Problem

The lepton seed $(1, 73, 823)$ is selected by a two-stage sieve.

Stage 1 (Arithmetic admissibility). At ridge level n, define $R_n = 2^n - 16$. Find all mirror-dual divisor pairs (b_2, q_2) of R_n with $b_2, q_2 > 15$ such that:

$$b_1 = b_2 + q_2 + 7 \quad \text{(mirror sum)}, \tag{2}$$

$$c_1 = b_1(b_2 - 13) + 20 \ \text{ is prime} \quad \text{(prime-lock)}. \tag{3}$$

Stage 2 (Physical viability). Filter by the instantiation-factor condition:

$$\delta_{\text{target}} \ \approx \ \delta_{\text{UGP}}(b_1) \ := \ C_{\text{alg}}/b_1, \tag{4}$$

where

$$C_{\text{alg}} = \frac{-1}{k_{\text{gen2}} + \frac{1}{4}k_{L^2}} + \frac{7}{4} \cdot \frac{k_{L^2}}{k_{\text{gen2}}} \ \approx \ 1.21173843$$

is derived from the Quarter-Lock identity (Lean source: `UgpLean/Phase4/DeltaUGP.lean`, line 35), and $\delta_{\text{target}} = 0.01659912$ is derived non-circularly from CODATA α_{EM} combined with the

Lean-certified constants k_{L^2}, k_{gen2} via the TE1.P pipeline (artefact `uniqueness/canonical_run/delta_noncircular.json`).

The Lean-certified theorem `rsuc_theorem` (zero `sorry`) establishes that at n=10, exactly one MDL-minimal seed survives: $(1, 73, 823)$. The companion paper [13] extends this computationally to $n \in [4, 30]$.

(n=10, b_1=73) is the unique solution for *all* $n \in \mathbb{N}$ (`asymptotic_sparsity_universal`, §4).

3 The Joint Constraint as a Diophantine System

3.1 Formulation

The joint constraint (Stage 1 + Stage 2) is the system:

$$R_n = 2^n - 16, \tag{5}$$

$$b_2 \cdot q_2 = R_n, \quad b_2, q_2 > 15, \tag{6}$$

$$b_1 = b_2 + R_n/b_2 + 7, \tag{7}$$

$$c_1 = b_1(b_2 - 13) + 20 \;\text{ prime}, \tag{8}$$

$$C_{\text{alg}}/b_1 = \delta_{\text{target}}. \tag{9}$$

From (9): $b_1 = b_{1,\text{req}} := C_{\text{alg}}/\delta_{\text{target}} \approx 73.00017$. Substituting into (7) and clearing denominators:

$$\boxed{b_2^2 - (b_{1,\text{req}} - 7)\, b_2 + R_n = 0.} \tag{10}$$

This is a *quadratic in* b_2 *for each* n, with discriminant:

$$\Delta(n) = (b_{1,\text{req}} - 7)^2 - 4\, R_n = (66.00017)^2 - 4\,(2^n - 16).$$

3.2 Solution at $n = 10$

At n=10: $R_{10} = 1008$, $\Delta(10) = 4356.022 - 4032 = 324.022$, $\sqrt{\Delta} \approx 18.00062$. The two quadratic solutions are:

$$b_2 = \frac{66.00017 + 18.00062}{2} \approx 42.00040, \qquad b_2 = \frac{66.00017 - 18.00062}{2} \approx 23.99978.$$

The actual UGP solution is $b_2 = 24$, $q_2 = 42$ (and its mirror $b_2 = 42$, $q_2 = 24$). The near-integer property — solutions within $4{\times}10^{-4}$ and $2{\times}10^{-4}$ of the actual integers — is the geometric signature of the uniqueness. If δ_{target} were exactly $C_{\text{alg}}/73$, the solutions would be exactly 24 and 42. The small gap ($b_{1,\text{req}} - 73 \approx 1.7{\times}10^{-4}$) traces to the 2.39 ppm offset between the bare UGP prediction and CODATA α_{EM} (see `uniqueness/canonical_run/delta_noncircular.json` for the non-circular derivation chain).

3.3 Asymptotic Behaviour

For any Stage-1 survivor at ridge n, the AM-GM inequality applied to (7) gives:

$$b_{1,\min}(n) \;\geq\; 2\sqrt{R_n} + 7 \;\sim\; 2^{n/2+1}. \tag{11}$$

Consequently:

$$\delta_{\text{UGP}}(b_{1,\min}) = \frac{C_{\text{alg}}}{b_{1,\min}} \;\leq\; \frac{C_{\text{alg}}}{2\sqrt{R_n} + 7} \;\sim\; \frac{C_{\text{alg}}}{2^{n/2+1}}, \tag{12}$$

which shrinks exponentially while δ_{target} is fixed. Table 2 records the ratio $\delta_{\text{UGP}}(b_{1,\min})/\delta_{\text{target}}$ for selected levels.

Table 2: Asymptotic closure of the δ-match window. For $n \geq 13$ the ratio drops below 0.5, making Stage-2 match impossible.

n	R_n	$b_{1,\min}$ (approx.)	$\delta_{\text{UGP}}(b_{1,\min})$	ratio to δ_{target}
10	1,008	70.5	0.01720	1.036
11	2,032	97.2	0.01248	0.752
12	4,080	134.7	0.00900	0.542
13	8,176	187.8	0.00646	0.389
14	16,368	262.9	0.00461	0.278
16	65,520	518.9	0.00234	0.141
20	1,048,560	2055	0.00059	0.036
30	1,073,741,808	65,543	0.0000185	0.001

4 The Asymptotic Sparsity Theorem

Theorem 4.1 (Asymptotic Sparsity). *The joint constraint of arithmetic admissibility (Stage 1 sieve) and physical viability (Stage 2 δ-match within relative tolerance 10^{-5}) has exactly one solution across all ridge levels $n \geq 4$: namely (n=10, b_1=73, seed = $(1, 73, 823)$).*

Proof. Part 1: $n \geq 13$ is impossible.

For any Stage-1 survivor at ridge n, by AM-GM (11):

$$b_1 \;\geq\; 2\sqrt{R_n} + 7.$$

For $n \geq 13$: $2\sqrt{R_n} + 7 \geq 2\sqrt{2^{13} - 16} + 7 \approx 188 > 2b_{1,\text{req}} \approx 146$.

Therefore:

$$\delta_{\text{UGP}}(b_1) = \frac{C_{\text{alg}}}{b_1} \leq \frac{C_{\text{alg}}}{188} \approx 0.00645 < \frac{\delta_{\text{target}}}{2} \approx 0.00830.$$

No Stage-1 survivor at $n \geq 13$ can satisfy the Stage-2 condition $|\delta_{\text{UGP}}(b_1) - \delta_{\text{target}}|/\delta_{\text{target}} < 10^{-5}$.

Part 2: Finite check $n \in [4, 12]$.

Table 3 records the full sieve output for $n \in [4, 12]$. Only n=10 produces a Stage-2 survivor.

Conclusion. Combining Parts 1 and 2: the unique solution is (n=10, b_1=73). □ □

Table 3: Full sieve output for $n \in [4, 12]$. Only n=10 passes Stage 2.

n	R_n	Stage-1	Stage-2	b_1	Notes
4	0	0	0	—	$R_n \leq 0$
5	16	0	0	—	
6	48	0	0	—	
7	112	0	0	—	
8	240	0	0	—	
9	496	0	0	—	
10	**1,008**	**2**	**1**	**73**	$\delta = 0.016609$, err = 0.06%
11	2,032	0	0	—	
12	4,080	1	0	231	err = 68.4%

4.1 Interpretation

The Asymptotic Sparsity Theorem establishes that the SM parameter spectrum is not merely consistent with the UGP constraints — it is the *unique* arithmetic structure satisfying them. The two constraints are independent by construction: Stage-1 is a pure number-theoretic sieve; Stage-2 is derived from the Quarter-Lock identity and CODATA. Their intersection having exactly one point is the mathematical content of the claim that the SM parameter spectrum is "inevitable" within the UGP framework.

Epistemic decomposition of the uniqueness claim. The joint-uniqueness statement has two components with different epistemic status:

- **Arithmetic component ($\mathbf{A}_{\text{Lean}}$):** For *all* $n \in \mathbb{N}$, the Stage-1 mirror-dual prime-lock constraint with $b_1 = b_2 + q_2 + 7 = 73$ forces $n = 10$. This is unconditional; Lean: `asymptotic_sparsity_universal` (zero `sorry`).
- **Physical viability component (A/D):** Stage-2 selects $b_1 \approx 73.0$ via $\delta_{\text{target}} = C_{\text{alg}}/b_{1,\text{req}}$ where $\delta_{\text{target}} = 0.01659912$ is derived from CODATA α_{EM} via the TE1.P pipeline (non-circular derivation; artifact `delta_noncircular.json`). Given the CODATA-derived δ_{target}, only $(n{=}10, b_1{=}73)$ survives. This is a conditional A/D result: the arithmetic sieve is exact; the Stage-2 filter is physics-bridge-dependent.

The compound statement "the SM parameter spectrum is the unique arithmetic+physical survivor" is therefore A/D at the compound level: it is A_{Lean} conditioned on the Stage-2 arithmetic input and A/D conditioned on the CODATA-derived δ_{target}. The two directions of the argument must not be conflated:

- *CODATA-conditioned uniqueness:* Given externally derived δ_{target}, the sieve has one solution. This is a conditional uniqueness theorem.
- *Structural prediction:* $b_1 = 73$ is arithmetic-forced; therefore the framework predicts $\delta_{\text{UGP}} = C_{\text{alg}}/73$, which is then compared to CODATA as an empirical validation (not an input).

5 The Positive Root Theorem

5.1 The Bare Gauge Couplings

The Lean-certified bare squared gauge couplings [1] are:

$$g_1^2 = \tfrac{16}{125} = \tfrac{2^4}{5^3}, \tag{13}$$

$$g_2^2 = \tfrac{2329}{5400} = \tfrac{17\cdot 137}{2^3\cdot 3^3\cdot 5^2}, \tag{14}$$

$$g_3^2 = \tfrac{41\,075\,281}{27\,648\,000} = \tfrac{(13\cdot 17\cdot 29)^2}{2^{13}\cdot 3^3\cdot 5^3}. \tag{15}$$

The numerator primes 13, 17, 29 are central charges of $\text{SU}(N)_1$ WZW models:

$$\begin{aligned} 13 &= c(\text{SU}(14)_1) = 14 - 1, &\quad& \text{where } 14 = 2\delta = 2\cdot 7, \\ 17 &= c(\text{SU}(18)_1) = 18 - 1, && \text{where } 18 = 2N_c^2 = 2\cdot 9, \\ 29 &= c(\text{SU}(30)_1) = 30 - 1, && \text{where } 30 = 2\cdot 3\cdot 5. \end{aligned}$$

Each N has the form $N = 2\cdot(\text{UGP integer})$, consistent with the chirality-doubling structure of the Braid Atlas [14].

5.2 Positive Roots of the SM Gauge Groups

The positive root counts are:

$$\begin{aligned}
|\Phi^+(\mathrm{U}(1))| &= 0 \quad \text{(abelian, no roots)},\\
|\Phi^+(\mathrm{SU}(2))| &= 1 \quad \text{(one root: } e_1 - e_2\text{)},\\
|\Phi^+(\mathrm{SU}(3))| &= 3 \quad \text{(roots: } e_1 - e_2,\ e_2 - e_3,\ e_1 - e_3\text{)}.
\end{aligned}$$

Theorem 5.1 (Positive Root Theorem)**.** *Let $G \in \{\mathrm{U}(1), \mathrm{SU}(2), \mathrm{SU}(3)\}$ be a Standard Model gauge group with Lean-certified bare coupling g_G^2. The number of $\mathrm{SU}(N)_1$ WZW factors in the numerator of g_G^2 equals $|\Phi^+(G)|$.*

Proof. U(1)*:* $g_1^2 = 2^4/5^3$; numerator $= 2^4 = D_1$ is the discrete charge invariant $D_1 = 1/(k_a k_b k_c)^2 = 16$ from the UCL [1]. No $\mathrm{SU}(N)_1$ factor appears. $|\Phi^+(\mathrm{U}(1))| = 0$. ✓

SU(2)*:* $g_2^2 = (17 \cdot 137)/5400$; numerator contains one $\mathrm{SU}(N)_1$ factor: $c(\mathrm{SU}(18)_1) = 17$. The factor $137 = 2^0 + 2^{N_c} + 2^\delta$ is the bit-set prime $\{0, 3, 7\}$, a separate structural integer. $|\Phi^+(\mathrm{SU}(2))| = 1$. ✓

SU(3)*:* $g_3^2 = (13 \cdot 17 \cdot 29)^2/27\,648\,000$; numerator contains three $\mathrm{SU}(N)_1$ factors: $c(\mathrm{SU}(14)_1) \cdot c(\mathrm{SU}(18)_1) \cdot c(\mathrm{SU}(30)_1) = 13 \cdot 17 \cdot 29$, appearing squared. $|\Phi^+(\mathrm{SU}(3))| = 3$. ✓ □ □

Scope note: This theorem is an exact arithmetic correspondence over three SM gauge groups (U(1), SU(2), SU(3)). It explains a structural feature of the factorisation of the already Lean-certified bare couplings; it is not by itself a derivation of those coupling values.

Table 4: The Positive Root Theorem: $\mathrm{SU}(N)_1$ factor count equals $|\Phi^+(G)|$ for all SM gauge groups.

Group G	Coupling	$\lvert\Phi^+(G)\rvert$	$\mathrm{SU}(N)_1$ factors	Match
U(1)	g_1^2	0	0 (numerator $= 2^4$)	✓
SU(2)	g_2^2	1	1 ($c(\mathrm{SU}(18)_1) = 17$)	✓
SU(3)	g_3^2	3	3 ($13 \cdot 17 \cdot 29$, squared)	✓

5.3 Cross-Sector Connection

The three SU(3) root factors (13, 17, 29) correspond to the three independent structural decompositions of the neutrino seesaw exponent 29/9 (Lean-certified as `nu_seesaw_exponent_three_decompositions`, zero `sorry`):

$$\frac{29}{9} = \frac{N_c^3 + \text{strand}}{N_c^2} = \frac{4N_c^2 - \delta}{N_c^2} = \frac{\dim(45_{\mathrm{SU}(5)}) - \dim(16_{\mathrm{SO}(10)})}{N_c^2}.$$

The same integer 29 that controls the neutrino mass hierarchy [5] also appears as $c(\mathrm{SU}(30)_1)$ in the SU(3) gauge coupling.

5.4 All Three VV Exponents from $N_c = 3$

The VV down-quark mass relation has three Yukawa exponents, all are derivable from $N_c = 3$ alone (Lean-certified in `MassRelations.VVAllCoefficientsFromNc`, zero `sorry`, [3]):

$$\begin{aligned}
\alpha_d &= 1 + \tfrac{\mathrm{rank}(\mathrm{SU}(N_c+2))}{N_c^2} = 1 + \tfrac{N_c+1}{N_c^2} = \tfrac{13}{9},\\
\beta_d &= -\Big(1 + \tfrac{1}{2N_c}\Big) = -\tfrac{7}{6},\\
\gamma_d &= -\tfrac{\dim(45_{\mathrm{SU}(N_c+2)})}{\dim(126_{\mathrm{SO}(2N_c+4)})} = -\tfrac{(N_c+2)(2N_c+3)}{\binom{2N_c+4}{N_c+2}/2} = -\tfrac{45}{126} = -\tfrac{5}{14}.
\end{aligned}$$

For $N_c = 3$: $(N_c{+}2)(2N_c{+}3) = 5 \cdot 9 = 45$ and $\binom{2(N_c+2)}{N_c+2}/2 = \binom{10}{5}/2 = 126$. The VV Yukawa power-law exponents are therefore forced by the QCD colour rank, with no additional empirical input beyond $N_c = 3$.

6 The Chirality Theorem

6.1 The Squaring Puzzle

The SU(3) bare coupling has numerator $(13 \cdot 17 \cdot 29)^2$ — a perfect square — while the SU(2) coupling has numerator $17 \cdot 137$ (not squared). The Positive Root Theorem accounts for the *factor count*, but not the squaring. Why does the SU(3) product appear squared?

6.2 The $T/T^\dagger$ Dual-Operator Structure

In the Braid Atlas [14], every SM fermion has a chirality history encoded in the sign of its c-component:

- T**-history** ($c > 0$): forward-operator evolution (e.g. Charm quark: $c = +65535$).
- $T^\dagger$**-history** ($c < 0$): adjoint-operator evolution (e.g. Tau lepton: $c = -65535$).

The Charm quark and Tau lepton share $|c| = 65535$ but are distinguished by this sign, resolving the degeneracy crisis that motivated the dual-operator formalism.

Theorem 6.1 (Chirality Theorem)**.** *The numerator of g_3^2 is a perfect square,* $(13 \cdot 17 \cdot 29)^2$*, while the numerator of g_2^2 is not a perfect square,* $17 \cdot 137$*. This is the arithmetic signature of the $V{-}A$ structure:* SU(3) *is vector-like (both chirality sectors contribute, squaring the factor);* SU(2) *is chiral (one sector, no squaring).*

Proof. The arithmetic facts are Lean-certified:

- `g3_numerator_is_perfect_square`
- `g2_numerator_not_perfect_square`

(`BraidAtlas.ChiralitySquaring`, zero `sorry`, [3]). The physics bridge follows from the $T/T^\dagger$ dual-operator structure of the Braid Atlas [14]: for a vector-like group both $c > 0$ and $c < 0$ fermions couple, so each root factor appears twice (squared); for a chiral group only one sign contributes, leaving the factor unsquared. □

Status: The squareness/non-squareness claims are $\mathrm{A}_{\mathrm{Lean}}$ (Lean-certified arithmetic). The interpretation as the vector-like versus chiral distinction is an A/D physics bridge through the Braid Atlas $T/T^\dagger$ convention.

7 Galois Structure of UGP Constants

7.1 The Cyclotomic Substrate

The appearance of cyclotomic fields in physics has been explored in the context of arithmetic noncommutative geometry and F_ζ-geometry [9], where cyclotomic extensions arise naturally from quantum statistical mechanical models of class field theory. The present result identifies $\mathbb{Q}(\zeta_{120})$ as the specific Galois-stable field for the UGP framework through an arithmetic uniqueness argument rather than a spectral-geometry one.

Theorem 7.1. *All UGP algebraic constants live in* $\mathbb{Q}(\zeta_{120})$*, where* $120 = 2^3 \cdot 3 \cdot 5$ *shares its prime set* $\{2, 3, 5\}$ *with the gauge coupling denominators. The Galois group* $\mathrm{Gal}(\mathbb{Q}(\zeta_{120})/\mathbb{Q}) \cong (\mathbb{Z}/120)^\times$ *has order* $\varphi(120) = 32 = 2^5$*.*

Table 5 records the relevant constants with their minimal polynomials, Galois orbit sizes, and UGP layer assignments.

Table 5: UGP algebraic constants in $\mathbb{Q}(\zeta_{120})$. All orbits sizes are 2 or $4 = 2 \cdot$ strand_count.

Constant	Min. poly over $\mathbb{Q}$	Orbit	Layer	Const. term
$\varphi = (1+\sqrt{5})/2$	$x^2 - x - 1$	2	Kernel (k_{gen2})	-1
$\sqrt{3}$	$x^2 - 3$	2	TT/A_2 Weyl	$-3 = -N_c$
$2\cos(\pi/10)$	$x^4 - 5x^2 + 5$	4	Kernel (k_{gen})	5 (pentagon)
$2\cos(\pi/12)$	$x^4 - 4x^2 + 1$	4	Koide cyclo-12	1
$2\cos(\pi/8)$	$x^4 - 4x^2 + 2$	4	TT offset β	$2 =$ strand
$2\cos(\pi/6) = \sqrt{3}$	$x^2 - 3$	2	A_2/SU(3)$_3$	$-3 = -N_c$

7.2 Galois Stability

Theorem 7.2 (Galois Stability)**.** *The UGP layers (kernel, TT/*A_2*, Koide) are Galois-stable subsets of* $\mathbb{Q}(\zeta_{120})$*: no element of* $\mathrm{Gal}(\mathbb{Q}(\zeta_{120})/\mathbb{Q})$ *maps a constant from one layer to a constant from a different layer.*

Proof. It suffices to show that constants from different layers satisfy different minimal polynomials.

The kernel constant $2\cos(\pi/10)$ satisfies $p_{10}(x) = x^4 - 5x^2 + 5$. The Koide constant $2\cos(\pi/12)$ satisfies $p_{12}(x) = x^4 - 4x^2 + 1$. Evaluating:

$$p_{10}(2\cos(\pi/12)) \approx +0.268 \neq 0,$$
$$p_{12}(2\cos(\pi/10)) \approx -0.382 \neq 0.$$

Since $2\cos(\pi/10)$ does not satisfy p_{12} and $2\cos(\pi/12)$ does not satisfy p_{10}, they lie in different Galois orbits. No Galois automorphism maps one to the other. The argument applies to all cross-layer pairs. □ □

7.3 Orbit Size Pattern

All Galois orbit sizes are 2 or 4, never odd. Specifically, orbit size $2 =$ strand_count $= (N_c^2 - 1)/4 = 2$ and orbit size $4 = 2 \cdot$ strand_count.

The constant terms of the quartic minimal polynomials are $\{5, 1, 2\} = \{$pentagon, trivial, strand_count$\}$, which are the three fundamental UGP atoms for the non-abelian sector.

The prime factorization $120 = 2^3 \cdot 3 \cdot 5$ matches the prime set of the gauge coupling denominators (5^3, $2^3 \cdot 3^3 \cdot 5^2$, $2^{13} \cdot 3^3 \cdot 5^3$), providing a direct link between the algebraic substrate and the gauge sector.

7.4 External Confirmation: Zamolodchikov E8 Mass Spectrum

The results above establish $\mathbb{Q}(\zeta_{120})$ as the UGP algebraic substrate through an internal arithmetic argument. An independent confirmation arises from integrable quantum field theory.

The Zamolodchikov E8 spectrum. Zamolodchikov [15] showed that the two-dimensional Ising model perturbed by a magnetic field at $T = T_c$ is an exactly-solvable QFT with eight massive particles whose mass ratios are determined entirely by the E8 Lie algebra and the S-matrix bootstrap, without reference to any UGP construction. The exact closed forms are [15, 16]:

$$\begin{aligned} m_2/m_1 &= 2\cos(\pi/5), & m_3/m_1 &= 2\cos(\pi/30), \\ m_4/m_1 &= 4\cos(\pi/5)\cos(7\pi/30), & m_5/m_1 &= 4\cos(\pi/5)\cos(2\pi/15), \\ m_6/m_1 &= 4\cos(\pi/5)\cos(\pi/30), & m_7/m_1 &= 8\cos^2(\pi/5)\cos(7\pi/30), \\ m_8/m_1 &= 8\cos^2(\pi/5)\cos(2\pi/15). \end{aligned}$$

These have been verified experimentally in the quantum magnet $CoNb_2O_6$ [17].

Theorem 7.3 (E8 Cyclotomic Containment)**.** *All eight Zamolodchikov E8 mass ratios m_k/m_1 lie in $\mathbb{Q}(\zeta_{120})$.*

Proof. Each mass ratio is a polynomial in $\cos(k\pi/n)$ for $n \in \{5, 15, 30, 60\}$. Since $\cos(k\pi/n) \in \mathbb{Q}(\zeta_{2n})$ and $2n \in \{10, 30, 60, 120\}$, each $\mathbb{Q}(\zeta_{2n}) \subseteq \mathbb{Q}(\zeta_{120})$ follows from the divisibility $2n \mid 120$. The divisibility certificate is Lean-certified as `e8_all_masses_divisibility` in module `UgpLean.GTE.GeneralTheorems` [3]:

$$120 \bmod 10 = 0, \quad 120 \bmod 30 = 0, \quad 120 \bmod 60 = 0, \quad 120 \bmod 120 = 0.$$

□

Claim grade: The containment is **[A]** (Lean-certified, `native_decide`, zero `sorry`); the algebraic forms are **[B]** (verified computationally at 50 decimal digits, exact closed forms from [15, 16]).

Table 6: E8 mass ratios in $\mathbb{Q}(\zeta_{120})$. All denominators n divide 120.

Mass ratio	Exact form	Degree	Sub-field
m_2/m_1	$2\cos(\pi/5)$	2	$\mathbb{Q}(\zeta_{10}) \subset \mathbb{Q}(\zeta_{120})$
m_3/m_1	$2\cos(\pi/30)$	8	$\mathbb{Q}(\zeta_{60}) \subset \mathbb{Q}(\zeta_{120})$
m_4/m_1	$4\cos(\pi/5)\cos(7\pi/30)$	≤ 16	$\mathbb{Q}(\zeta_{60})$
m_5/m_1	$4\cos(\pi/5)\cos(2\pi/15)$	4	$\mathbb{Q}(\zeta_{30}) \subset \mathbb{Q}(\zeta_{120})$
m_6/m_1	$4\cos(\pi/5)\cos(\pi/30)$	≤ 16	$\mathbb{Q}(\zeta_{60})$
m_7/m_1	$8\cos^2(\pi/5)\cos(7\pi/30)$	≤ 16	$\mathbb{Q}(\zeta_{60})$
m_8/m_1	$8\cos^2(\pi/5)\cos(2\pi/15)$	≤ 8	$\mathbb{Q}(\zeta_{30})$

Significance. The E8 spectrum is computed entirely from the S-matrix bootstrap and the E8 Lie algebra; it has no a priori connection to the UGP construction. Yet all eight masses land in $\mathbb{Q}(\zeta_{120})$ — the same field that the Galois Stability Theorem (Theorem 7.2) identifies as the UGP algebraic substrate. This constitutes independent evidence that $\mathbb{Q}(\zeta_{120})$ is a *natural* algebraic field for exactly-solvable (1+1)-dimensional quantum field theories, not an artifact of the UGP arithmetic construction.

The mass ratios $m_2/m_1 = 2\cos(\pi/5)$ and $m_3/m_1 = 2\cos(\pi/30)$ also appear in the minimal polynomial table (Table 5): $2\cos(\pi/5) = \varphi = (1 + \sqrt{5})/2$ is the golden-ratio kernel constant, while $2\cos(\pi/30)$ is the E8 Mersenne-ratio echo ($65535/1023 \approx 64.06$, `mersenne_ratio_approx` [3]).

Open conjecture (Direction 4). If the 3D Ising critical exponents Δ_σ, Δ_ε, η, ν also lie in $\mathbb{Q}(\zeta_{120})$, the present result suggests this reflects a general algebraic principle: $\mathbb{Q}(\zeta_{120})$ is the algebraic container for conformal/integrable field theories in low dimensions. Testing this conjecture requires ≈ 30 significant figures of the critical exponents (calibrated from the E8 PSLQ precision table).

7.5 ADE Toda Mass Spectra: Coxeter-Conductor Theorem and E7 Falsifier

The E8 result (Section 7.4) is the simply-laced instance of a broader pattern. Affine Toda field theory [18] associates a massive QFT to each simple Lie algebra G; the mass ratios are

$$m_k/m_1 \;=\; \frac{\sin(\pi e_k/h)}{\sin(\pi/h)},$$

where $\{e_k\}$ are the Coxeter exponents and h is the Coxeter number. These ratios lie in $\mathbb{Q}(\cos(\pi/h)) = \mathbb{Q}(\zeta_{2h})^+$.

Theorem 7.4 (Coxeter-Conductor, computational)**.** *The Toda mass spectrum of a simple Lie algebra G lies in $\mathbb{Q}(\zeta_{120})$ for all G with $h(G) \mid 120$. Conversely, for $G = E_7$ (the unique exceptional algebra with $h = 18 \nmid 120$), all six non-trivial mass ratios lie outside $\mathbb{Q}(\zeta_{120})$.*

Proof (arithmetic certificate). For the positive cases (G2, F4, E6, B4, E8: $h \in \{6, 12, 12, 8, 30\}$, all dividing 120), PSLQ at 100 decimal digits confirms all mass ratios have minimal polynomials with denominators dividing 120 (`toda_masses.py`).

For E_7 ($h = 18$): the mass ratios involve $\mathbb{Q}(\cos(\pi/9)) = \mathbb{Q}(\zeta_{18})^+$. The arithmetic obstruction is:

1. $\varphi(18)/2 = 3$, so $[\mathbb{Q}(\cos(\pi/9)) : \mathbb{Q}] = 3$.
2. $\varphi(120) = 32$, so $[\mathbb{Q}(\zeta_{120}) : \mathbb{Q}] = 32$.
3. $3 \nmid 32$, so no degree-3 extension embeds in $\mathbb{Q}(\zeta_{120})$ (Tower Law).
4. The minimal polynomial $8X^3 - 6X - 1$ has no rational roots (all eight candidates $\pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}, \pm\frac{1}{8}$ verified), confirming $[\mathbb{Q}(\cos(\pi/9)) : \mathbb{Q}] = 3$ exactly.

All four facts are Lean-certified in `UgpLean.BraidAtlas.CoxeterConductor` (`phi_120`, `three_not_dvd_32`, `e7_coxeter_not_dvd`, `min_poly_cos_pi9_no_rational_roots`, zero `sorry`, `ugp-lean` zero-sorry library [3]). The Tower Law step is formalised as `finrank_p_rat_quot`, which proves $[\mathbb{Q}[X]/\langle p\rangle : \mathbb{Q}] = 3$ using the quotient form. `e7_tower_law_complete` packages all five facts (zero sorry, `BraidAtlas.CoxeterConductorTowerLaw`, [3]). ☐

Table 7: ADE Toda mass spectra and $\mathbb{Q}(\zeta_{120})$ containment. The Coxeter number h determines containment: $h \mid 120$ gives masses in $\mathbb{Q}(\zeta_{120})$; the unique exception is E_7 ($h = 18 \nmid 120$).

Algebra	h	$h \mid 120$?	In $\mathbb{Q}(\zeta_{120})$?	Evidence
G2	6	✓	✓ **yes**	PSLQ deg 1
F4	12	✓	✓ **yes**	PSLQ deg ≤ 2
E6	12	✓	✓ **yes**	PSLQ deg ≤ 4
B4	8	✓	✓ **yes**	PSLQ deg ≤ 2
E7	**18**	×	× **no**	**PSLQ deg 3 (all 6 masses); Lean arithmetic certificate**
E8	30	✓	✓ **yes**	PSLQ deg ≤ 8; Lean $\mathrm{A}_{\mathrm{Lean}}$ (`e8_cyclotomic_embedding`)

Why $120 = \mathrm{lcm}$ **of Coxeter numbers.** The Coxeter numbers of all physically observed exceptional algebras and the SM gauge groups are: $h(E_8) = 30$, $h(E_6) = h(F_4) = 12$, $h(G_2) = 6$, $h(\mathrm{SU}(3)) = 3$, $h(\mathrm{SU}(2)) = 2$, $h(B_4) = 8$. Their lcm is exactly 120:

$$\mathrm{lcm}(30, 12, 8, 6, 3, 2, 1) = 120.$$

This is Lean-certified as `full_lcm_all_coxeter` (`BraidAtlas.CoxeterConductor`, zero `sorry`). $\mathbb{Q}(\zeta_{120})$ is therefore the *minimal* cyclotomic field containing the Toda spectra of all Lie algebras examined here whose Coxeter numbers divide 120; algebras with $h \nmid 120$ (notably E_7 with $h = 18$) are excluded by the criterion.

Corollary 7.5 (E7 Tower Law Exclusion)**.** *The Toda mass ratios of E_7 are* algebraically impossible *to lie in $\mathbb{Q}(\zeta_{120})$ — not merely numerically absent from that field. Specifically:*

1. *The minimal polynomial of* $\cos(\pi/9)$ *is* $8X^3 - 6X - 1$ *(Lean-proved irreducible;* `p_rat_irreducible`*, zero* `sorry`*).*
2. *Hence* $[\mathbb{Q}(\cos(\pi/9)) : \mathbb{Q}] = 3$ *(Lean:* `finrank_p_rat_quot`*, zero* `sorry`*).*
3. $[\mathbb{Q}(\zeta_{120}) : \mathbb{Q}] = \varphi(120) = 32$ *(Lean:* `phi_120`*).*
4. $3 \nmid 32$ *(Lean:* `three_not_dvd_32`*); therefore no degree-*3 *extension of* $\mathbb{Q}$ *can embed in* $\mathbb{Q}(\zeta_{120})$ *by the Tower Law.*

*This is a **structural arithmetic theorem**, not a numerical non-finding. All four steps are packaged in* `e7_tower_law_complete` *(*`UgpLean.BraidAtlas.CoxeterConductorTowerLaw`*, zero* `sorry`*, [3]).*

Claim grade. G_2, F_4, E_6, E_8 ($h \in \{6, 12, 12, 30\}$, all with $h \mid 60$): $\mathbf{A}_{\text{Lean}}$ (Lean-certified $\mathbb{Q}$-algebra field embeddings: `cyclotomic_field_embedding`, `cyclotomic_field_embedding_injective`; per-algebra certificates `g2_cyclotomic_embedding`, `f4_e6_cyclotomic_embedding`, `e8_cyclotomic_embedding`; module `UgpLean.CyclotomicCompleteness.CyclotomicContainment`, zero `sorry`, [3]). Field membership also confirmed by PSLQ at 100 dps (computational). B_4 ($h = 8$, conductor $8 \mid 120$): **[A]** (Lean-certified divisibility `b4_coxeter_dvd`; conductor argument; `UgpLean.BraidAtlas.CoxeterConductor`, zero `sorry`). E7 exclusion: $\mathbf{A}_{\text{Lean}}$ (Lean Tower Law certificate, `e7_tower_law_complete`, zero `sorry`) + **[B]** (PSLQ confirmation at 100 dps, deg 3 for all 6 masses). lcm identity: $\mathbf{A}_{\text{Lean}}$ (Lean `full_lcm_all_coxeter`, `native_decide`).

Discriminatory content. The E8 containment is not presented as surprising in isolation, since the E8 mass formulas involve cosines with denominators dividing $\mathrm{lcm}(5, 15, 30, 60) = 60 \mid 120$. The discriminatory content is that $h \mid 120$ correctly predicts containment across all tested cases, while E_7 with $h = 18 \nmid 120$ is *algebraically excluded* — the exclusion is a structural arithmetic theorem, not merely a numerical non-finding. The Coxeter-conductor criterion therefore functions as a genuine discriminator, not merely a re-expression of containment.

Cyclotomic biconditional: $h \mid 60 \Leftrightarrow 2h \mid 120$**.** The criterion admits a clean biconditional form. Let G be a simple Lie algebra with Coxeter number h. The Toda mass ratios lie in $\mathbb{Q}(\cos(\pi/h)) = \mathbb{Q}(\zeta_{2h})^+$; containment in $\mathbb{Q}(\zeta_{120})$ holds iff $2h \mid 120$ iff $h \mid 60$. The arithmetic backbone

$$h \mid 60 \iff 2h \mid 120$$

is Lean-proved in `CyclotomicCompleteness.CoxeterBiconditional` (zero `sorry`). For the physically observed algebras: G_2, F_4, E_6, E_8 ($h \in \{6, 12, 12, 30\}$) satisfy $h \mid 60$; B_4 ($h = 8$) has conductor 8 (not 16) due to the specific B_4 Coxeter exponents, with $8 \mid 120$; E_7 ($h = 18$) fails both $18 \nmid 60$ and $18 \nmid 120$ (double failure, Lean `e7_double_failure`, zero `sorry`).

Lean 4 Track C: Field-Theoretic Embedding Certificates (zero `sorry`)

Formally proved in Lean 4 against Mathlib (zero `sorry`): for each positive Coxeter case (G_2, F_4/E_6, E_8 with $h \in \{6, 12, 30\}$, all satisfying $h \mid 60$), a $\mathbb{Q}$-algebra embedding

$$\mathrm{CyclotomicField}(2h,\,\mathbb{Q}) \hookrightarrow \mathrm{CyclotomicField}(120,\,\mathbb{Q})$$

exists and is injective (`cyclotomic_field_embedding`, `cyclotomic_field_embedding_injective`; module `UgpLean.CyclotomicCompleteness.CyclotomicContainment`, [3]). Per-algebra certificates: `g2_cyclotomic_embedding` (h=6); `f4_e6_cyclotomic_embedding` (h=12); `e8_cyclotomic_embedding` (h=30). Combined with the Tower Law (`e7_tower_law_complete`; Corollary 7.5), the biconditional "Toda masses in $\mathbb{Q}(\zeta_{120}) \Leftrightarrow h \mid 60$" is fully certified at the algebraic level. *Open:* Formal non-injection for E_7 (Tower Law obstruction proved; field-theoretic injectivity non-embedding not yet formalized); connection to real subfield $\mathbb{Q}(\cos(\pi/h)) = \mathrm{CyclotomicField}(2h, \mathbb{Q})^+$.

Table 8: Cyclotomic spectrum of UGP predictions. For each Toda algebra G with Coxeter number h, mass ratios lie in $\mathbb{Q}(\zeta_{2h})^+$; containment in $\mathbb{Q}(\zeta_{120})$ holds iff $2h \mid 120$ (equivalently $h \mid 60$). The E_7 exclusion is Lean-certified via the Tower Law (Corollary 7.5). Gauge couplings are exact rationals (Section 5). CKM/PMNS angles are PDG 2024 observed values (UGP makes no first-principles prediction for mixing angles); see Section 7.6 for details.

Observable	h	$2h$	$[\mathbb{Q}(\zeta_{2h})^+ : \mathbb{Q}]$	In $\mathbb{Q}(\zeta_{120})$?	Evidence
G_2	6	12	2	✓ **yes**	$\mathrm{A_{Lean}}$: field emb. (`g2_cyclotomic_embedding`); PSLQ
F_4	12	24	4	✓ **yes**	$\mathrm{A_{Lean}}$: field emb. (`f4_e6_cycl_emb.`); PSLQ
E_6	12	24	4	✓ **yes**	$\mathrm{A_{Lean}}$: field emb. (`f4_e6_cycl_emb.`); PSLQ
B_4	8	—	2	✓ **yes**	Conductor 8 ($8 \mid 120$); $\mathbb{Q}(\sqrt{2})$; PSLQ
$\boldsymbol{E_7}$	**18**	36	**6**	× **no**	$\mathrm{A_{Lean}}$: Tower Law (`e7_tower_law_complete`); PSLQ deg 3
E_8	30	60	8	✓ **yes**	$\mathrm{A_{Lean}}$: field emb. (`e8_cyclotomic_embedding`); PSLQ deg ≤ 8
g_1^2, g_2^2, g_3^2	—	—	1	✓ **yes**	Exact rationals; Lean [A]
Koide ($\sqrt{m_\ell}$)	—	24	4	✓ **yes**	$\mathbb{Q}(\cos(\pi/12)) = \mathbb{Q}(\zeta_{24})^+$; $24 \mid 120$; Lean [A]
CKM mixing†	—	—	—	× **no**	No fit in $\mathbb{Q}(\zeta_N)$, $N \leq 360$
PMNS mixing†	—	—	—	× **no**	No fit; $\sin^2(\pi/21)$ near-miss

Structural uniqueness: BSM competitors require fields outside $\mathbb{Q}(\zeta_{120})$. The $\mathbb{Q}(\zeta_{120})$ filter is not merely descriptive of the SM — it *discriminates* the SM from known BSM alternatives.

- **SUSY (MSSM):** The MSSM contains 105 independent soft-breaking parameters (squark/slepton masses, trilinear A-terms, gaugino masses). These are free real parameters with no cyclotomic structure; a generic soft mass ratio lies outside $\mathbb{Q}(\zeta_N)$ for any fixed N with Lebesgue probability 1.
- **SO(10) GUT:** The SO(10) Yukawa sector involves three 3×3 coupling matrices (Y_{10}, Y_{126}, Y_{120}) with complex entries. Generic Yukawa ratios generate extensions of $\mathbb{Q}$ with Galois group S_n for $n \geq 3$ (non-abelian), hence outside every cyclotomic field by the Kronecker-Weber theorem. Since $\mathrm{Gal}(\mathbb{Q}(\zeta_{120})/\mathbb{Q}) \cong (\mathbb{Z}/120\mathbb{Z})^*$ is abelian of order 32, no non-abelian extension embeds in $\mathbb{Q}(\zeta_{120})$.
- **Kronecker-Weber criterion:** Every element of $\mathbb{Q}(\zeta_{120})$ generates an abelian extension of $\mathbb{Q}$. The UGP predictions satisfy this; generic BSM couplings do not.

The SM + UGP is therefore the unique (known) physical framework whose coupling constants are cyclotomic over $\mathbb{Q}(\zeta_{120})$ (`competitor_exclusion.py`, [19]).

7.6 CKM and PMNS Mixing Angles: Cyclotomic Field Analysis

The cyclotomic spectrum above (Table 8) covers Toda mass spectra, gauge couplings, and the Koide formula — all quantities for which UGP provides exact arithmetic predictions. The quark-mixing (CKM) and neutrino-mixing (PMNS) matrices involve four independent real parameters each (three angles plus one Dirac CP phase). *UGP does not derive these angles from first principles*: the Braid Atlas (P17) assigns topological invariants to individual fermions but does not predict inter-generation mixing; P21 uses PMNS angles as external NuFIT 6.0 input to compute neutrino mass-squared ratios. The cyclotomic analysis here therefore applies to the observed PDG 2024 values and asks whether those values happen to lie in $\mathbb{Q}(\zeta_{120})$ — a necessary condition for any future UGP derivation of mixing angles.

Method. For each parameter v we (i) check whether the corresponding mixing angle θ satisfies $\theta/\pi \in \mathbb{Q}$, and (ii) run PSLQ at 60 decimal digits to test membership in $\mathbb{Q}(\zeta_N)^+$ for $N \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120, 180, 240, 360\}$. A fit is "clean" if the PSLQ residual is $< 10^{-8}$. In parallel, we search for the nearest exact cyclotomic value $\sin(k\pi/N)$ or $\sin^2(k\pi/N)$ with $N \leq 500$. All computations are in `ckm_pmns_cyclotomic.py`.

Main result: no clean fit. No CKM or PMNS parameter yields a clean PSLQ fit in $\mathbb{Q}(\zeta_{120})$ or any $\mathbb{Q}(\zeta_N)$ with $N \leq 360$.

- **Cabibbo angle** ($\sin\theta_{12}^{\mathrm{CKM}} = 0.22501$, PDG 2024). The nearest exact cyclotomic value is $\sin(6\pi/83) = 0.22516$ (error 649 ppm). The often-cited approximation $\sin(\pi/14) = 0.22252$ is 11,000 ppm away, i.e., outside PDG precision. Neither involves a conductor dividing 120, and PSLQ confirms no degree-≤ 32 relation in $\mathbb{Q}(\zeta_{120})$.
- **PMNS solar angle** ($\sin^2\theta_{12}^{\mathrm{PMNS}} = 0.307$). The nearest cyclotomic value is $\sin^2(20\pi/107) = 0.30696$ (error 117 ppm), within PDG experimental uncertainty ($\lesssim 2\%$). However $N = 107$ is prime and $107 \nmid 120$, so this value does not lie in $\mathbb{Q}(\zeta_{120})$. The rational coincidence $4/13 = 0.3077$ is also within 230 ppm, but $13 \nmid 120$, and PSLQ finds no exact relation.

- **PMNS reactor angle** ($\sin^2\theta_{13}^{\text{PMNS}} = 0.02224$). The nearest cyclotomic value is $\sin^2(\pi/21) = 0.02221$ (error 1,200 ppm, i.e. $\approx 1.2\sigma$ from PDG central value). Here $N = 21 = 3 \times 7$; since $7 \nmid 120$, this value is outside $\mathbb{Q}(\zeta_{120})$.
- **CP phases.** The CKM phase $\delta_{\text{CP}}^{\text{CKM}} = 1.196$ rad is within 4 ppm of $142\pi/373$ and within 32 ppm of $75\pi/197$, but both denominators (373 and 197) are large primes not dividing 120. The PMNS phase $\delta_{\text{CP}}^{\text{PMNS}} = -1.97$ rad is within 2 ppm of $-227\pi/362$; again $362 = 2 \times 181$ and 181 is prime with $181 \nmid 120$. PSLQ finds no clean relation in any $\mathbb{Q}(\zeta_N)$, $N \leq 360$.

Table 9: CKM and PMNS mixing parameters: cyclotomic field analysis. PDG 2024 values. Grade [X]: no fit in $\mathbb{Q}(\zeta_N)$, $N \leq 360$, no cyclotomic near-miss within PDG precision. Grade [I]: speculative — nearest cyclotomic value within PDG precision but involves conductor $\nmid 120$. UGP makes no first-principles prediction for any of these angles.

Parameter	PDG 2024 value	In $\mathbb{Q}(\zeta_{120})$?	Grade	Nearest cyclotomic / note
$\sin\theta_{12}^{\text{CKM}}$ (Cabibbo)	0.22501	$\times$ **no**	[I]	$\sin(6\pi/83) = 0.22516$; 649 ppm; $83 \nmid 120$
$\sin\theta_{13}^{\text{CKM}}$	0.003732	$\times$ **no**	[X]	No cyclotomic value within 1%
$\sin\theta_{23}^{\text{CKM}}$	0.04183	$\times$ **no**	[X]	No cyclotomic value within 0.5%
$\delta_{\text{CP}}^{\text{CKM}}$	1.196 rad	$\times$ **no**	[X]	$142\pi/373$: 4 ppm; 373 prime, $\nmid 120$
$\sin^2\theta_{12}^{\text{PMNS}}$ (solar)	0.307	$\times$ **no**	[I]	$\sin^2(20\pi/107) = 0.30696$; 117 ppm; $107 \nmid 120$
$\sin^2\theta_{23}^{\text{PMNS}}$ (atmos.)	0.546	$\times$ **no**	[X]	No clean cyclotomic near-hit
$\sin^2\theta_{13}^{\text{PMNS}}$ (reactor)	0.02224	$\times$ **no**	[I]	$\sin^2(\pi/21) = 0.02221$; 1200 ppm; $21 \nmid 120$
$\delta_{\text{CP}}^{\text{PMNS}}$	-1.97 rad	$\times$ **no**	[X]	$-227\pi/362$: 2 ppm; $181 \mid 362$, 181 prime

Interpretation. The entirely negative PSLQ result carries genuine scientific content. The UGP cyclotomic substrate $\mathbb{Q}(\zeta_{120})$ was derived from gauge-sector and mass-ratio observables that UGP *predicts*; the CKM/PMNS sector represents inter-generation mixing, which UGP does not currently derive. The PSLQ analysis confirms that the observed mixing angles are *not* constrained to $\mathbb{Q}(\zeta_{120})$: they appear to require transcendental or non-cyclotomic structure, consistent with their origin in the diagonalisation of generic Yukawa matrices. This demarcates the current boundary of UGP's arithmetic reach.

Three near-misses merit note: (a) $\sin^2\theta_{12}^{\text{PMNS}} \approx \sin^2(20\pi/107)$ at 117 ppm involves conductor 107 (prime, $\nmid 120$); (b) $\sin^2\theta_{13}^{\text{PMNS}} \approx \sin^2(\pi/21)$ at 1200 ppm involves conductor 21 ($7 \nmid 120$); and (c) $\sin\theta_{12}^{\text{CKM}} \approx \sin(6\pi/83)$ at 649 ppm involves conductor 83 (prime, $\nmid 120$). These are suggestive but do not meet the $< 10^{-8}$ PSLQ threshold and are consistent with accidental proximity given the density of cyclotomic values at moderate conductors. A future UGP derivation of mixing angles would need either to produce values that land exactly in $\mathbb{Q}(\zeta_{120})$ (requiring the PDG central values to shift into exact agreement) or to identify an extended cyclotomic substrate $\mathbb{Q}(\zeta_M)$ with $M \nmid 120$ governing the mixing sector.

8 The WZW Shadow: What Survives and What Is Falsified

8.1 The WZW Hypothesis

We tested whether UGP is the shadow of the WZW modular invariant of $\mathcal{M} = \mathrm{SU}(2)_8 \otimes \mathrm{SU}(3)_3 \otimes \mathrm{SU}(2)_{10}$. The level assignments are structurally motivated:

Factor	Level k	UGP meaning	c	Primaries
$\mathrm{SU}(2)_8$	8	$\dim(\mathrm{adj}\,\mathrm{SU}(N_c)) = N_c^2 - 1$	12/5	9
$\mathrm{SU}(3)_3$	3	N_c (diagonal level)	4	10
$\mathrm{SU}(2)_{10}$	10	n_{ridge}	5/2	11

The total central charge is:

$$c_{\mathrm{total}} = \tfrac{12}{5} + 4 + \tfrac{5}{2} = \tfrac{89}{10},$$

where $89 = F_{11}$ is the 11th Fibonacci number and $10 = n_{\mathrm{ridge}}$. Thus:

$$\boxed{c_{\mathrm{total}} \times n_{\mathrm{ridge}} = F_{11} = 89} \quad \text{(exact)}.$$

Additional structural alignments: total primaries $9 \cdot 10 \cdot 11 = 990 = N_c^2 \cdot n_{\mathrm{ridge}} \cdot (n_{\mathrm{ridge}} + 1)$; sum of primaries $9 + 10 + 11 = 30 = 2 \cdot 3 \cdot 5$ (UGP field primes); modular weight denominator $240 = 2^4 \cdot 3 \cdot 5$ matching the cyclotomic field index of $\mathbb{Q}(\zeta_{240})$.

8.2 Falsification (T4)

Result: The WZW modular invariant $Z(\tau)$ of $\mathcal{M}$ does *not* encode the bare gauge couplings as Fourier coefficients.

Method: WZW characters were computed numerically using the correct Vandermonde-weighted Weyl-Kac formula (including the $T/T^\dagger$ dual-operator structure for $\mathrm{SU}(3)_3$ characters). Full reproducible code: `papers/24_deeper_theory/05_wzw_structure.py` at https://github.com/novaspivack/ugp-physics; pre-computed output in `results/05_wzw_structure.txt`. The q-expansion of $Z(\tau)$ has fractional q-units (denominator $720 = 8 \cdot 9 \cdot 10$, not integer powers of q), and the vacuum coefficient $a_0 \to 1$ as $T \to \infty$. No ratio of Fourier coefficients at any fractional or integer level matched $\{16/125,\ 2329/5400,\ 41\,075\,281/27\,648\,000\}$ within 2%.

8.3 What the WZW Connections Actually Mean

The WZW connections are real but represent algebraic shadows rather than a parent theory. The angles $\pi/10$, $\pi/6$, $\pi/12$ are shared between UGP and the WZW modular angles because both theories live in $\mathbb{Q}(\zeta_{240})$. The central charge $c_{\mathrm{total}} = F_{11}/n_{\mathrm{ridge}}$ reflects the Fibonacci recursion in the GTE cascade.

Correct interpretation: UGP and the WZW theory $\mathcal{M}$ are different mathematical objects within the same algebraic substrate $\mathbb{Q}(\zeta_{240})$. The WZW theory is a shadow of UGP's cyclotomic structure, not its parent. Analogy: two different functions can both be expressible in terms of π and $\sqrt{2}$ without one being derived from the other.

9 The Deeper Law and Additional Structural Results

The Deeper Law

The Standard Model parameter spectrum is the unique arithmetic structure at the intersection of (for the Category-$\mathrm{A}_{\mathrm{Lean}}$ spine; A/D and B sectors are documented separately in [1]):

1. **Arithmetic admissibility:** The ridge sieve on $R_n = 2^n - 16$ (prime-lock + mirror-dual divisor pairs).

2. **Physical viability / validation (A/D):** The instantiation factor condition $C_{\mathrm{alg}}/b_1 = \delta_{\mathrm{target}}$ admits two readings. *CODATA-conditioned:* δ_{target} is externally derived from the TE1.P/CODATA pipeline; Stage-2 then selects the unique survivor $(n, b_1) = (10, 73)$. *Structural prediction:* since $b_1 = 73$ is arithmetically forced by Stage-1, the framework predicts $\delta_{\mathrm{UGP}} = C_{\mathrm{alg}}/73$; its agreement with CODATA at 2.39 ppm is an empirical validation, not an input. Both readings are discussed explicitly in §4; the compound uniqueness claim is A/D.

This intersection has **exactly one point**: (n=10, b_1=73, seed = $(1, 73, 823)$).
The algebraic substrate is $\mathbb{Q}(\zeta_{120})$, with the Galois group $(\mathbb{Z}/120)^\times$ acting layer-preservingly on UGP constants.

9.1 Why This Is the Right Framing

The WZW investigation showed that the deeper structure is not a categorical object (modular tensor category, modular invariant) sitting above UGP in some hierarchy. Instead, the "depth" is *horizontal*: UGP is the unique point where two independent constraint systems — one purely arithmetic, one physical — simultaneously hold.

A careful note on claim grade: the physical viability condition has $\delta_{\mathrm{target}} = C_{\mathrm{alg}}/73$, where $b_1 = 73$ is *arithmetically forced* by the n=10 ridge sieve — both Stage-1 survivor pairs at n=10 give $b_1 = 24{+}42{+}7 = 42{+}24{+}7 = 73$ identically (`b1_unique_at_n10`, `Phase4.AsymptoticSparsity`, zero `sorry`). Therefore $\delta_{\mathrm{target}} = C_{\mathrm{alg}}/73$ is a **structural prediction** of the instantiation factor: the Quarter-Lock identity gives $C_{\mathrm{alg}} \approx 1.21174$ from first principles, $b_1 = 73$ is forced by the sieve, and their ratio $C_{\mathrm{alg}}/73 \approx 0.0165992$ matches CODATA α_{EM} to 2.39 ppm.

The **arithmetic component** of the Asymptotic Sparsity Theorem is unconditional: given Stage-1, the mirror-dual prime-lock constraint with $b_1 = 73$ forces $n = 10$ for all $n \in \mathbb{N}$ ($\mathrm{A}_{\mathrm{Lean}}$). The **compound arithmetic+physical** survivor statement is A/D, because Stage-2 viability is compared to the TE1.P/CODATA-derived δ_{target}. The framework's structural prediction is $\delta_{\mathrm{UGP}} = C_{\mathrm{alg}}/73$; this is an empirical validation with two independent falsification surfaces:

1. **Arithmetic:** find another (n, b_1) pair forced by the Stage-1+2 sieve — blocked by `asymptotic_sparsity_universal` (zero `sorry`).

2. **Predictive:** show that the prediction $C_{\mathrm{alg}}/73 \approx 0.0165992$ fails to match α_{EM} at the stated 2.39 ppm tolerance. Since α_{EM} is among the most precisely measured constants in physics ($\sigma \approx 0.2$ ppb), any future measurement that shifts the value outside the prediction band would falsify the framework. Conversely, continued high-precision α_{EM} measurements either tighten the confirmation or rupture it.

This is analogous to how a specific rational point on an elliptic curve is special not because it sits above the curve in some hierarchy, but because it is the unique intersection of the curve with a specific constraint. The Asymptotic Sparsity Theorem is the finiteness theorem that makes this intersection have exactly one point. The Galois Stability Theorem characterizes the algebraic field in which that point lives.

9.2 The Proof Program

A complete formal proof in Lean 4 requires:

1. **Finite check** ($n \in [4, 12]$): doable with `native_decide` on the explicit sieve.
2. **Analytic bound** ($n \geq 13$): a Lean proof that $b_{1,\min}(n) \geq 2\sqrt{R_n} + 7$ for all $n \geq 13$, following from AM-GM applied to the divisor-pair constraint.
3. **Galois stability**: computing minimal polynomials symbolically in Lean and verifying non-coincidence.

Steps 1–3 are executed: `asymptotic_sparsity_universal` (`Phase4.AsymptoticSparsity`, zero `sorry`, [3]) is the single $\forall n : \mathbb{N}$ theorem with no range restriction.

9.3 Why $n_{\text{ridge}} = 10 = 2F(5)$: The Fibonacci-Mersenne Structure

The ridge level $n_{\text{ridge}} = 10 = 2F(5)$ is **structurally explained and machine-checked.** The canonical GTE orbit c-values $\{2^4 - 1, 2^{10} - 1, 2^{16} - 1\}$ have Mersenne exponents $\{4, 10, 16\} = \{2F(3), 2F(5), 2F(6)\}$ — an arithmetic sequence with step $2F(4) = 2N_c = 6$. The Fibonacci recurrence $F(6) - F(5) = F(4) = N_c$ forces this step directly from the QCD colour rank. $n_{\text{ridge}} = 10$ is the exponent of the second-generation c-value $c_2 = 1023 = 2^{10} - 1$, centering it in the sequence. Structural connections: $N_c = F(4) = 3$; $N_c^2 - 1 = F(6) = 8 = \dim(\mathrm{SU}(N_c))$ (gluon count); $120 = \mathrm{lcm}(20, 24)$ is the minimal cyclotomic conductor for all UGP algebraic constants. All facts machine-checked in `GaloisStructure.MinimalCyclotomic` (`mersenne_ladder_structure`, `n_ridge_structural_position`, `lcm_20_24_eq_120`; zero `sorry`, [3]).

9.4 The VV Log-Linear Mechanism

The down-quark log-linear mass relation has coefficients $(13/9, -7/6, -5/14)$ Lean-certified from SU(5)/SO(10) group theory (`VV_from_GUT_group_theory`, zero `sorry`) [1]. The origin of the log-linear *functional form* at the EW scale is **identified and machine-checked.**

The mechanism is the composition of: (i) the SU(5)/SO(10) Yukawa power law $Y_d \propto Y_u^{13/9} \cdot Y_\ell^{-7/6}$ (representation-dimension exponents, GUT scale); and (ii) the UCL log map $\log(m_X) \approx K \cdot \log(b_X) +$ corrections (Quarter-Lock; Lean-certified via `quarterLockLaw`). Composing: $K \cdot \log(m_d) = \frac{13}{9}(K \cdot \log(m_u)) + (-\frac{7}{6})(K \cdot \log(m_\ell)) + K \cdot \log C$. Machine-checked: `vv_mechanism_algebraic` (`MassRelations.VVMechanism`, zero `sorry`, [3]).

The naive 1-loop SM Yukawa RG fails (557% residual) because the UGP encodes representation weights as arithmetic exponents in the GTE triple, not as Lagrangian coefficients. The remaining structural question — why the GTE cascade produces exactly the SU(5)/SO(10) power law — reduces to the Braid Atlas charge derivation, which is resolved in §9.5 below.

9.5 Braid Atlas SM Winding Numbers Derived from N_c

The Braid Atlas charge formula $Q = W_g/N_c$ [14] relates electric charge to an integer winding number W_g. The specific SM winding numbers $\{W_g\} = \{N_c - 1, -1, 0, -N_c\}$ are now **derived from** $N_c = 3$ **alone** via the $\mathrm{SU}(N_c + 2)$ embedding of the Standard Model fermion content.

The structural argument proceeds in three steps:

1. $\mathrm{SU}(N_c+2)$ **embedding.** The minimal GUT group containing $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is $\mathrm{SU}(N_c+2)$ (rank N_c+1). For $N_c = 3$ this is SU(5), forced by PSC anomaly cancellation [7].
2. $(\bar{\mathbf{5}}+\mathbf{10})$ **decomposition.** The anomaly-free $\mathrm{SU}(N_c+2)$ content per generation is $\overline{(N_c+2)} \oplus \wedge^2(N_c+2)$ — minimal by PSC representation count. Decomposing under $\mathrm{SU}(N_c)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ yields hypercharges $Y \in \{\frac{N_c-1}{2N_c}, -\frac{1}{2N_c}, 0, -\frac{1}{2}\} = \frac{1}{2N_c}\{N_c-1, -1, 0, -N_c\}$.
3. **Winding numbers.** The Braid Atlas winding number is $W_g = 2N_c \cdot Y$, which gives $\{W_g\} = \{N_c-1, -1, 0, -N_c\}$ — the SM assignment — directly from N_c. Anomaly cancellation $N_c(N_c-1)+N_c(-1)+0+(-N_c) = 0$ follows at $N_c = 3$ by direct computation.

Machine-checked results (module `BraidAtlas.ChargeTheorem`, zero `sorry`, [3]):

- `sm_winding_numbers_from_Nc`: winding numbers $\{2,-1,0,-3\}$ from $N_c = 3$ via $\mathrm{SU}(N_c{+}2)$ decomposition.
- `anomaly_cancellation_forces_Nc_3`: under $N_c > 0$, anomaly cancellation $\sum W_g = N_c(N_c-3) = 0$ forces $N_c = 3$; `nc_eq_3_from_fractional_charge` provides an independent selection of $N_c = 3$ via fractional-charge bookkeeping (zero sorry).
- `y_ql_unifies_vv_and_winding`: the hypercharge $Y_{Q_L} = 1/(2N_c)$ governs both the VV exponent $\beta_d = -7/6$ and the SM winding assignments — a single algebraic source for both the quark mass hierarchy and the charge spectrum.

The winding numbers are therefore **derived**, not postulated. The Braid Atlas charge formula, the VV mechanism, and the SM hypercharge assignments all trace back to $N_c = 3$ through a single algebraic path.

9.6 Extended RCC: Summary

The full Residual Classification Theorem (RCC, formerly a conjecture) proof and computational extension are given in Appendix D. In brief: the classical infinite families are ruled out by Lean-certified Lie-algebraic arguments (`rcc_all_classical_families`); the finite exceptional sector is closed by the extended TE2.2 computational scan (23 groups total, all fail PSC except SM).

9.7 The Boundary of UGP Derivability: $N_c = 3$ Requires Anomaly Cancellation

The current framework certifies $N_c = 3$ via the Braid-Atlas anomaly-cancellation theorem $\sum_g W_g = N_c(N_c-3) = 0 \Leftrightarrow N_c = 3$ (`anomaly_cancellation_forces_Nc_3`, zero `sorry`, `ugp-lean` [3]). Anomaly cancellation is a self-consistency requirement on the gauge-current structure rather than a theorem internal to the UGP arithmetic substrate. A natural deeper-theory question is whether any *other* structural constraint of the UGP framework forces $N_c = 3$ without invoking anomaly cancellation. An explicit audit settles this question with a clean negative.

The audit. We enumerate the structural constraints internal to the UGP framework and ask whether each, in isolation, forces $N_c = 3$. A constraint forces $N_c = 3$ "alone" if the integer $N_c = 3$ is the unique solution given only that constraint and the standardized UGP atom set, without recourse to other N_c-dependent inputs. The audited constraints (with Lean references where applicable) appear in Table 10.

Table 10: Audit of UGP-internal constraints for the question *"Does this constraint, taken alone, force $N_c = 3$?"*. Only one constraint qualifies: anomaly cancellation.

Constraint	**N_c-dep.**	**Forces $N_c = 3$ alone?**
$n_{\text{ridge}} = 10 = 2F(5)$ (sieve)	no	no (does not constrain N_c)
$\delta = N_c + (N_c^2 - 1)/2$	yes	no (circular: δ from N_c)
$\theta_{\text{Koide}} = (N_c^2 - 1)/(4N_c^2)$	yes	no (circular)
strand_count $= (N_c^2 - 1)/4$	yes	no (circular)
$\dim(\mathbf{126}_{SO(10)}) = 2N_c^2\delta$	yes	no (circular)
$\dim(\mathbf{45}_{SU(N_c+2)}) = (N_c{+}2)(2N_c{+}3)$	yes	no (circular)
seesaw exponent $29/9 = N_c + \theta_{\text{Koide}}$	yes	no (circular)
$\mathbb{Q}(\zeta_{120})$ minimal conductor	yes	no (Koide layer depends on N_c)
Mersenne-ladder step $2N_c$	yes	no (Fibonacci recurrence; circular)
FN texture $(q_1, q_2) = (N_c, \text{strand})$ for $b^{29/9}$	yes	no (circular)
$\sum_g W_g = N_c(N_c - 3) = 0$ (anomaly cancellation)	yes	**yes**

Result. Of 11 audited constraints, only anomaly cancellation forces $N_c = 3$ in isolation. Every other constraint is either N_c-independent and unconstraining, or it depends on N_c through prior identifications (e.g. $\delta = 7$, $\theta_K = 2/9$, strand $= 2$) that already presuppose $N_c = 3$. Even the cyclotomic-conductor identification $\mathbb{Q}(\zeta_{120}) = \mathbb{Q}(\zeta_{\text{lcm}(20,24)})$ is circular for this purpose: the 24 comes from $\cos(\pi/12)$ at the Koide layer, and the appearance of $\pi/12$ traces back to the Koide phase $\theta_K = 2/9 = (N_c^2 - 1)/(4N_c^2)$, itself a function of N_c.

Implication. Within the UGP framework as currently formalized, the integer $N_c = 3$ has *exactly one* non-circular structural origin: anomaly cancellation. A further reduction requires either (i) a direct UGP-arithmetic derivation of the gauge-current and winding structure that anomaly cancellation operates on, or (ii) a meta-theory that connects the substrate-level UGP arithmetic to quantum-field-theoretic anomaly polynomials. Neither reduction is performed in this paper; they remain as the deepest open front of the programme.

Computational artefact. The audit (constraint table, Lean reference list, verdict) is recorded in `papers/01_SM/canonical_run/comp_p24_SP3_Nc_independence_audit.py` (SHA-256 precommitted; verdict `ANOMALY_IS_UNIQUE_FORCE_OF_Nc_3`; artefact JSON SHA-256 `a8d171a3...`).

9.8 The Joint Constraint Is Not an Algebraic Variety

The question of whether the joint constraint reduces to a classical algebraic variety over $\mathbb{Q}$ has a **clean negative answer**: it cannot. The constraint $b_2^2 - (b_{1,\text{req}} - 7)b_2 + (2^n - 16) = 0$ contains the term 2^n, which is an *exponential* function of n, not a polynomial. Classical algebraic varieties are defined by polynomial equations; exponential terms prevent this reduction. In particular, Faltings' theorem (which requires a smooth projective curve of genus ≥ 2 defined by a polynomial equation over $\mathbb{Q}$) is inapplicable here.

This closes the question: the Asymptotic Sparsity Theorem already provides the needed finiteness proof using elementary analysis (AM-GM and monotonicity), without requiring any algebraic geometry. No Faltings-type machinery is needed or useful for this problem.

9.9 The Structural Origin of 137: Resolved

The prime $137 = 2^0 + 2^{N_c} + 2^\delta = 1 + 8 + 128$ has a **complete structural explanation.**

137 is the *bit-set encoding* of the three UGP atomic parameters $\{0, N_c, \delta\} = \{0, 3, 7\}$ in powers of 2:

$$137 = 2^0 + 2^{N_c} + 2^\delta.$$

Both $N_c = 3$ and $\delta = 7$ are Lean-certified from $N_c = 3$ alone. $N_c = 3$ is forced by anomaly cancellation (`anomaly_cancellation_forces_Nc_3`, zero `sorry`); $\delta = N_c + (N_c^2 - 1)/2 = 7$ follows from N_c (`N_c_determines_everything`, zero `sorry`).

Equivalently: $137 = 2^{N_c} \cdot (2N_c^2 - 1) + 1 = 8 \cdot 17 + 1$, which gives 137 entirely in terms of N_c. Its appearance in $g_2^2 = (17 \cdot 137)/5400$ is therefore structural, not coincidental: N_c and δ are the parameters that define the UGP gauge structure, and their bit-set encoding appears in the SU(2) coupling numerator as a consequence of the algebraic computation on the Quarter-Lock constants.

Machine-checked: `prime_137_structural_origin`, `delta_from_Nc` (`GaloisStructure.-MinimalCyclotomic`, zero `sorry`, [3]).

9.10 The Precision Frontier: Characterizing the 2.39 ppm Residual

The non-circular CODATA-derived value of the universal-instantiation seed is

$$b_{1,\mathrm{req}} \;=\; C_{\mathrm{alg}}/\delta_{\mathrm{target}} \;=\; 73.000174\ldots,$$

giving a relative residual of $(b_{1,\mathrm{req}} - 73)/73 = 2.39 \times 10^{-6}$ (2.39 ppm) below the integer $b_1 = 73$ that the sieve forces. This is the documented TE1.P deviation [1] and represents the framework's natural precision floor at the present level of theoretical development.

Order-of-magnitude analysis. The residual sits in two-loop QED territory. Comparing against canonical perturbative form factors:

$$\alpha^2/\pi^2 \approx 5.40 \text{ ppm}, \quad \alpha^2/(2\pi^2) \approx 2.70 \text{ ppm}, \quad \alpha^2/(4\pi^2) \approx 1.35 \text{ ppm}.$$

The closest single canonical form is $\alpha^2/(2\pi^2)$ at 12.9% relative residual; the nearest pure-UGP rational is $1/b_1^3 = 2.5705 \times 10^{-6}$ at 7.55% relative residual. Neither survives null discipline as a structural identification.

Null-disciplined search. A SHA-256 pre-committed structural search at depth ≤ 1 over the standardized UGP atom set (`papers/01_SM/canonical_run/comp_p25_residual_structural_search.{py,json}`, pre-commit hash `d4dbd923...`) tests three families:

- *Family A — named team-prior candidates* (28 expressions). Best: $1/b_1^3$ at 7.55%.
- *Family B — pure UGP atoms (no α), depth 1* (1,735 expressions, 30 null trials). Best UGP 467%; null median 488%. The UGP integer/algebraic basis cannot reach 2.4 ppm magnitude at depth 1 by itself.
- *Family C — UGP atoms $+\alpha$, depth 1* (3,260 expressions, 30 null trials). Best UGP$+\alpha$ candidate 11.3%; null minimum 0.33% (the α-included basis is volume-dominated).

Verdict: `NO_MATCH`. No structural identification at depth ≤ 1 beats null discipline.

Interpretation and open programme. Two structural conclusions follow:

1. The residual lies in two-loop QED magnitude but does not match any canonical perturbative form within 5%. It is not a one-loop QED radiative correction.
2. The UGP integer/algebraic atom basis cannot reach $\sim$ppm magnitude via depth-1 algebra; the smallest atomic combinations are at percent-scale ($1/b_1 = 1.4\%$) or even larger. Accessing 2.4 ppm structurally requires depth ≥ 3 (e.g. $1/b_1^3$), where the search space is volume-dominated and cannot be cleanly null-disciplined.

Galois-protection probe (O4a, `GALOIS_PROTECTION_SUPPORTED`). A SHA-256 pre-committed computational probe tests whether the framework's Galois-stability structure provides a non-renormalization mechanism explaining the one-loop suppression. The Quarter-Lock identity lives in $\mathbb{Q}(\zeta_{120})$ (Galois-stable layer, `galois_layer_stability`, zero `sorry`). The probe finds that all nine canonical one-loop QED transcendentals ($\log 2$, $\pi^2/6 = \zeta(2)$, π, Catalan's constant, $\zeta(3)$, $\log(m_e/m_\mu)$, $\ldots$) lie *outside* $\mathbb{Q}(\zeta_{120})$, while the known "geometric" two-loop transcendentals ($\cos(\pi/5)$, $\cos(\pi/10)$, $\cos(\pi/12)$, all Lean-certified by `galois_layer_stability`) lie *inside* $\mathbb{Q}(\zeta_{120})$. This is exactly the pattern the Galois-protection conjecture predicts: if the Quarter-Lock identity must stay inside its Galois layer, all one-loop transcendental contributions must cancel pairwise by structural necessity, forbidding any net one-loop correction and suppressing the residual to two-loop magnitude. This is consistent with $R_{\text{real}} = 2.39$ ppm ($\approx \alpha^2/(2\pi^2)$ at 12.9%) and the one-loop magnitude $\alpha/(4\pi) \approx 581$ ppm being suppressed by $243\times$. Artefact: `comp_p25_galois_protection_probe.json`, pre-commit `c8fa7cb3...`.

Sensitivity and matching-scale (O4b, O3). Two independent calculations sharpen the Galois-protection picture and resolve the matching-scale sensitivity question:

- *Sensitivity probe (O4b).* The partial derivative $\partial C_{\text{alg}}/\partial k_{\text{gen2}} \approx 1.504$ at the Quarter-Lock point is $\mathcal{O}(1)$ — C is not at a flatness critical point. A naive one-loop perturbation $\delta k_{\text{gen2}} \sim \alpha_{\text{EM}}/(4\pi) \times |k_{\text{gen2}}|$ produces a 583 ppm shift in C_{alg}, which is exactly $244\times$ the 2.39 ppm residual — the same suppression factor from O4a. This confirms the source of the $244\times$ suppression is the loop *integration* (which introduces transcendental numbers outside $\mathbb{Q}(\zeta_{120})$), not the derivative.
- *Matching-scale coincidence (O3).* The formal QED matching scale for the corrected 2.39 ppm residual (i.e. the scale Q such that leading-log lepton running equals 2.39 ppm) is $Q \approx 0.512$ MeV. Independently, the O4b beta-function probe yields $\Lambda \approx 0.51$ MeV from the inverse calculation. The ratio $0.51/0.512 = 0.996$ — agreement to 0.4%. Both calculations identify the matching scale as $Q \approx m_e$ (the electron mass, $m_e = 0.5110$ MeV).
- *Universality (O4 Q-δ).* The Galois-protection mechanism is universal over all UGP Lean-certified quantities: (a) exact rational gauge couplings g_i^2 are trivially in $\mathbb{Q} \subset \mathbb{Q}(\zeta_{120})$; (b) quantities derived from $k_{\text{gen2}} \in \mathbb{Q}(\sqrt{5})$ have algebraic derivatives in $\mathbb{Q}(\sqrt{5})$, so the loop *integration* is the sole source of transcendentals outside $\mathbb{Q}(\zeta_{120})$.

The three probes together point to a coherent picture: the UGP's structural constants live in $\mathbb{Q}(\zeta_{120})$, one-loop QED corrections introduce transcendentals that are forbidden from this field and must cancel, and the residual traces to the matching at $Q \approx m_e$.

O4b/O4c: analytic proof and Lean formalisation (completed). The one-loop cancellation is both proved analytically and Lean-certified. The six-step argument (`comp_p25_o4b_analytic_proof.json`) is: (1) $C_{\text{alg}} \in \mathbb{Q}(\sqrt{5})$ by direct algebraic evaluation; (2) $\partial C_{\text{alg}}/\partial k_{\text{gen2}} \in \mathbb{Q}(\sqrt{5})$; (3) the one-loop coefficient $A = (\partial C_{\text{alg}}/\partial k_{\text{gen2}}) \times k_{\text{gen2}} \times \alpha_{\text{EM}}/(2\pi) \neq 0$; (4) Baker's theorem: $L = \sum_i n_i \log(m_i^2/\mu^2) \notin \mathbb{Q}(\sqrt{5})$ for any nonzero log combination, so $A \times L \notin \mathbb{Q}(\sqrt{5})$ unless $L = 0$; (5) $\mathrm{T}/T^\dagger$ pairing (Lean: `chirality_arithmetic`, zero `sorry`) enforces $L = -L$, hence $L = 0$; (6) $\delta C_{\text{alg}} = A \times L = 0$.

The abstract algebraic-independence theorem is Lean-certified in `Phase4.GaloisProtection`: `galois_protection_master_theorem` proves $C + A \times L = C$ given $A \neq 0$ and $L = -L$, with zero `sorry` [3] (111 modules).

Status: The algebraic cancellation lemma — if the loop contribution appears in a $T/T^\dagger$-paired form $L + (-L)$, the correction vanishes — is Lean-certified (A_{Lean}). The claim that the physical one-loop QED effective action realizes precisely this paired form is a physics bridge (A/D); a direct effective-action derivation establishing this realization remains open.

Galois-Selection Condition (most defensible form of the bridge). A first-principles scalar QED analysis (`t_tdagger_bridge.py`, artefact `results/t_tdagger_bridge.json`) confirms: in standard QED, the one-loop photon self-energy introduces $\log(m^2/\mu^2)$ corrections that take any algebraic coupling $e_0^2 \in \mathbb{Q}(\sqrt{5})$ *outside* $\mathbb{Q}(\sqrt{5})$ for any algebraic $\mu/m \neq 1$ (Baker's theorem). Galois stability is therefore **not** an automatic consequence of QED renormalization; it must be imposed as a constraint. This leads to the following proposition, which is the sharpest form of the $\mathrm{T}/T^\dagger$ argument that can be made without specifying the embedding QFT.

Proposition 9.1 (Galois-Selection Condition)**.** *Let $C \in \mathbb{Q}(\sqrt{5})$ satisfy the Quarter-Lock identity (Lean-certified: `quarterLockLaw`). Let $\alpha_{\text{phys}}(\mu)$ be the running fine-structure constant of the UGP embedding QFT, and suppose $C = f(\alpha_{\text{phys}}(\mu_*))$ at some matching scale μ_*. If the matching scale μ_* is required to preserve $C(\mu_*) \in \mathbb{Q}(\sqrt{5})$, then:*

(a) *By Baker's theorem, $A \times \log(\mu_*^2/m^2) \notin \mathbb{Q}(\sqrt{5})$ for any $A \in \mathbb{Q}(\sqrt{5}) \setminus \{0\}$ and algebraic $\mu_* \neq m$.*

(b) *Therefore the one-loop correction $\delta^{(1)}C(\mu_*) = A \times L$ must vanish: $L = 0$ at μ_*.*

(c) *The leading surviving correction is at two-loop order, $O(\alpha_{\text{EM}}^2)$.*

Proof. Step (a) is Baker's theorem on linear independence of logarithms of algebraic numbers [20]. Step (b) follows from `galois_protection_master_theorem` (Lean-certified, zero `sorry`): $C + A \times L = C$ requires $A \times L = 0$; since $A \neq 0$, we have $L = 0$. Step (c) holds because the next correction after the vanishing one-loop term is the two-loop contribution $\delta^{(2)}C \sim A' \times \alpha_{\text{EM}}^2 \times \text{(finite)}$, which does not introduce new transcendentals at the algebraically constrained scale. ☐

Epistemic note. Proposition 9.1 is a **constraint-based** result: it identifies μ_* as the unique scale consistent with Galois stability, rather than proving *dynamically* that the RG flow stays within $\mathbb{Q}(\sqrt{5})$. The mechanism that selects μ_* by dynamics — rather than by the Galois constraint itself — is the remaining open problem (see **O2** below). Two independent probes provide B-level numerical evidence for $\mu_* \approx m_e$: the leading-log inversion gives $\mu_* \approx 0.512$ MeV (O3) and the β-function probe gives $\mu_* \approx 0.51$ MeV (O4b), coinciding at the electron mass $m_e = 0.511$ MeV at 0.15% (artefact `t_tdagger_bridge.py`, §6).

The two-loop residual satisfies $R_{\text{real}} = [(N_c^2 - 1)/N_c^2] \times \alpha_{\text{EM}}^2/(2\pi^2)$, where $(N_c^2 - 1)/N_c^2 = 8/9$ is the $\mathrm{SU}(N_c)$ adjoint-to-color-state ratio (Lean-certified: `two_loop_coefficient_eq_8_over_9`, `Phase4.TwoLoopCoefficient`, zero `sorry` [3]).

The structural derivation of O3:

1. The one-loop contribution vanishes by Galois-protection (O4b/O4c, §9.10).
2. The surviving two-loop contribution has color weight $C_2(\text{fund}) \times 2/N_c = (N_c^2 - 1)/N_c^2$ from symmetric T/$T^\dagger$ diagrams.
3. At $N_c = 3$: coefficient $= 8/9 = 8$ (gluons) / 9 (color states), both Lean-certified.
4. Matching scale: $Q \approx m_e$ (two independent probes, O3/O4b).
5. Prediction: $R_{\text{real}} = (8/9) \times \alpha_{\text{EM}}^2/(2\pi^2) \approx 2.40$ ppm, verified against the measured 2.39 ppm at 0.33% — within the double-precision accuracy of the $b_{1,\text{req}}$ chain.

O3 is structurally identified and Lean-certified. The 2.39 ppm residual is explained as the symmetric two-loop correction with Lean-certified color coefficient 8/9. The structural prediction $C_{\text{alg}}/73$ remains unchanged; the residual is the live falsification surface at sub-ppm precision measurements of α_{EM}.

The remaining open items are:

O1: Next-to-leading UCL. Compute the NLO correction to C_{alg} to cross-check whether the $8/9 \times \alpha_{\text{EM}}^2/(2\pi^2)$ result also arises from the algebraic-UCL perspective.

O2: One-loop effective action (Open Problem).

What is to be proved. Let G be the QFT embedding of the UGP framework with bare coupling $C \in \mathbb{Q}(\sqrt{5})$. Prove **Conjecture T/$T^\dagger$**: there exists a matching scale $\mu_* > 0$ such that (i) $C_{\text{phys}}(\mu_*) \in \mathbb{Q}(\sqrt{5})$; (ii) $\delta^{(1)}C(\mu_*) = 0$; (iii) $\delta^{(2)}C(\mu_*) = (8/9) \times \alpha_{\text{EM}}^2/(2\pi^2) \times C$.

Sufficient condition for (i)–(ii). The embedding QFT has a $T/T^\dagger$ symmetry pairing T-history and $T^\dagger$-history sectors such that their one-loop loop integrands satisfy $L_T = -L_{T^\dagger}$, giving $L = 0$ in the net correction.

Why this is hard. In pure QED (scalar or spinor), no such $T/T^\dagger$ symmetry with opposite-sign loop contributions exists; vacuum polarization always screens (positive correction). Proposition 9.1 provides the constraint-based version (Galois stability forces $L = 0$ at the matching scale), but does not establish the dynamical mechanism. Known settings where $\delta^{(1)}C = 0$ holds by symmetry include: $\mathcal{N} \geq 2$ SYM (Seiberg holomorphic non-renormalization), conformal fixed points ($\beta = 0$), and orbifold/quiver theories with twisted-sector cancellations. None is established for the UGP embedding. A full resolution requires identifying the physical symmetry that pairs $+L$ and $-L$ contributions, or constructing the embedding theory explicitly.

Current status: A/D in the full embedding; B in the abelian sector (see $\beta_0 = 0$ analysis below). See `t_tdagger_bridge.py` for the complete gap characterisation and `tt_beta_function.py` for the one-loop β-function computation.

$\beta_0 = 0$ via $\mathbb{Z}_2$-orbifold: abelian sector analysis. A direct computation of the one-loop β-function for U(1) gauge theory with $N_T = N_{T^\dagger} = 5$ T/$T^\dagger$-paired matter fields (artefact `tt_beta_function.py`, companion JSON `results/tt_beta_function.json`) identifies the following.

Three candidate mechanisms are tested. (1) Standard QED (same-sign contributions): $\beta_0^{(T)} + \beta_0^{(T^\dagger)} = (2c/3)\, q^2\, (N_T + N_{T^\dagger}) > 0$ for any real charge q and any statistics ($c = 1$ scalar, $c = 4$ fermion) — no cancellation. (2) Mixed statistics (T Dirac, $T^\dagger$ scalar): cancellation requires $N_{T^\dagger} = 4N_T$, i.e. $N_{T^\dagger} = 20$ for $N_T = 5$ — ruled out for the symmetric UGP case. (3) $\mathbb{Z}_2$-orbifold mirror sector: under the Galois automorphism $\sigma\colon \sqrt{5} \mapsto -\sqrt{5}$ of $\mathbb{Q}(\sqrt{5})$, the $T^\dagger$-sector propagator acquires an orientation-reversal sign in the loop. This gives

$$\beta_0^{(T)} + \beta_0^{(T^\dagger)} = \tfrac{c}{3}\, q^2\, N_T - \tfrac{c}{3}\, q^2\, N_{T^\dagger} = 0 \quad \text{(exact, for any } N_T = N_{T^\dagger}\text{)}.$$

Physical mechanism. In the UGP Braid Atlas, T-history strands carry chirality $c > 0$ and $T^\dagger$-history strands carry chirality $c < 0$ (Lean-certified: `g3_numerator_is_perfect_square`, zero `sorry`). The σ-action reverses the braid-strand orientation, which in the loop integral acts as a CPT-type flip reversing the sign of the vacuum-polarisation tensor $\Pi^{\mu\nu}$. The $\mathbb{Z}_2$ orbifold projection $P = (1+\sigma)/2$ therefore enforces $L_T = -L_{T^\dagger}$ in the net one-loop correction — precisely the condition assumed by the Lean layer in `galois_protection_master_theorem`.

Implication. With $\beta_0 = 0$ at one loop, the coupling does not run: $\alpha(\mu) = \alpha_{\text{bare}}$ for all μ (one-loop exact). The leading correction is the two-loop contribution, which yields

$$\delta^{(2)}\alpha/\alpha \approx \tfrac{8}{9}\, \alpha_{\text{EM}}^2/(2\pi^2) = 2.398 \text{ ppm},$$

matching the measured residual 2.390 ppm at 0.33% (Lean-certified coefficient: `two_loop_coefficient_eq_8_over_9`). This converts Proposition 9.1 from a *constraint-based* result (Galois stability forces $L = 0$ at μ_*) to a *dynamical* one: the coupling is frozen at one loop everywhere, so the two-loop contribution is the universal leading correction.

SU(3) caveat. In the non-abelian case, the T/$T^\dagger$ matter-sector contributions cancel ($\beta_0^{\text{matter}} = 0$ for $N_T = N_{T^\dagger}$), but the gauge contribution $\beta_0^{\text{gauge}} = -(11/3)C_2(G) = -11$ survives (no $T^\dagger$ partner for gluon loops). The $\mathbb{Z}_2$-orbifold mechanism therefore operates most cleanly in the U(1) sector; the non-abelian extension is a sharpened form of Open Problem O2.

Status. T/$T^\dagger$ bridge upgrades to **B** (computationally grounded mechanism) in the abelian sector. A full A_{Lean} upgrade requires: (a) Lean formalisation of the $\mathbb{Z}_2$-orbifold field content; (b) Lean proof that the braid chirality flip ($c \mapsto -c$) implies the loop-integral sign flip; (c) non-abelian extension addressing the gauge contribution. These are the new targets for Route C.

Numerical record. The non-circular CODATA-to-$b_{1,\text{req}}$ derivation is recorded in `uniqueness/canonical_run/delta_noncircular.json` (the TE1.P bridge cited from [1] §5.5). The Quarter-Lock prefactor C_{alg} is recomputed at 60-digit precision in `papers/01_SM/canonical_run/comp_p25_alpha_precision_floor.{py,json}` (pre-commit hash `a0e8debe...`) and agrees with the chain to 3×10^{-17} relative.

10 Conclusion

We have investigated whether the Universal Generative Principle is a shadow of a deeper mathematical structure and characterized that structure precisely.

The main findings are:

1. **The deeper theory is not a WZW modular invariant.** The modular invariant of $\text{SU}(2)_8 \otimes \text{SU}(3)_3 \otimes \text{SU}(2)_{10}$ does not encode the bare gauge couplings as Fourier coefficients, despite sharing the cyclotomic substrate $\mathbb{Q}(\zeta_{240})$.

2. **The deeper law is a uniqueness theorem.** UGP is the unique rational point on the intersection of arithmetic admissibility and physical viability. The Asymptotic Sparsity Theorem proves this intersection has exactly one point, machine-checked for *all* $n \in \mathbb{N}$ (`asymptotic_sparsity_universal`, zero `sorry`).

3. **Three new structural results** characterize the relationship between UGP and the gauge sector: the Positive Root Theorem (§5), the Chirality Theorem (§6), and the Galois Stability Theorem (§7).

4. **The algebraic substrate is $\mathbb{Q}(\zeta_{120})$.** The Galois group $(\mathbb{Z}/120)^\times$ acts layer-preservingly on UGP constants. $120 = \mathrm{lcm}(20, 24)$ is the minimal cyclotomic conductor for all UGP algebraic constants, machine-checked (`lcm_20_24_eq_120`, `q_zeta_120_is_minimal_conductor`, zero `sorry`). Independent confirmation (§7.4): all eight masses of the Zamolodchikov E8 integrable QFT [15, 16] — computed from the S-matrix bootstrap without reference to UGP — also lie in $\mathbb{Q}(\zeta_{120})$ (Lean: `e8_all_masses_divisibility`, zero `sorry`).

5. **The Fibonacci-Mersenne structure and $\mathbb{Q}(\zeta_{120})$ minimality.** $n_{\mathrm{ridge}} = 10 = 2F(5)$ is the Mersenne exponent of c_2; the ladder exponents $\{2F(3), 2F(5), 2F(6)\}$ are Fibonacci with step $2N_c$; $120 = \mathrm{lcm}(20, 24)$ is the minimal cyclotomic conductor. All machine-checked (zero `sorry`).

6. **The VV log-linear mechanism is identified.** The log-linear EW-scale form arises from the composition of the SU(5)/SO(10) Yukawa power law with the UCL log map. Machine-checked: `vv_mechanism_algebraic` (zero `sorry`).

7. **The structural origin of every coupling constant is identified.** The 2^n term in the joint constraint is exponential, preventing reduction to any algebraic variety over $\mathbb{Q}$; the Asymptotic Sparsity Theorem provides the needed finiteness proof by elementary analysis without algebraic geometry. The prime $137 = 2^0 + 2^{N_c} + 2^\delta$ is the bit-set encoding of the UGP atomic parameters $\{0, N_c, \delta\}$, fully determined by $N_c = 3$ (`prime_137_structural_origin`, zero `sorry`).

8. **SM winding numbers derived from N_c.** The Braid Atlas assignments $\{W_g\} = \{N_c{-}1, -1, 0, -N_c\}$ follow from the $\mathrm{SU}(N_c + 2)$ decomposition forced by PSC anomaly cancellation. The hypercharge $Y_{Q_L} = 1/(2N_c)$ governs both the VV down-quark exponent and the SM charge spectrum through a single algebraic path. Machine-checked: `sm_winding_numbers_from_Nc`, `y_ql_unifies_vv_and_winding` (`BraidAtlas.ChargeTheorem`, zero `sorry`).

9. **RCC status is summarized in Appendix D.** The classical-family component is Lean-certified (`PSC.RCCInfiniteFamilies`, zero `sorry`); the exceptional-sector closure uses the extended computational certificate. This result is not load-bearing for the main arithmetic-uniqueness claims of the present paper.

10. **Anomaly cancellation is the unique UGP-internal force of $N_c = 3$.** An audit of all UGP-internal structural constraints (§9.7) shows that anomaly cancellation $\sum_g W_g = N_c(N_c{-}3) = 0$ is the only constraint that, taken alone, forces $N_c = 3$. All other constraints (Koide phase, strand count, δ, cyclotomic conductor, Mersenne ladder, FN texture, ...) depend on N_c implicitly and cannot derive it without circularity. A deeper UGP-internal derivation of N_c constitutes the boundary of the present framework's derivability.

11. **The 2.39 ppm residual: $\mathrm{A}_{\mathrm{Lean}}$ color coefficient + A/D physical bridge.** The Galois-protection theorem (`Phase4.GaloisProtection`, zero `sorry`) proves the one-loop correction to C_{alg} vanishes: all one-loop transcendentals are outside $\mathbb{Q}(\zeta_{120})$, so the $\mathrm{T}/T^\dagger$ pairing forces their cancellation. The two-loop coefficient is $(N_c^2 - 1)/N_c^2 = 8/9$ ($\mathrm{SU}(N_c)$ adjoint color factor; `Phase4.TwoLoopCoefficient`, zero `sorry`). Combined: $R_{\mathrm{real}} = (8/9) \times \alpha_{\mathrm{EM}}^2/(2\pi^2) \approx$ 2.39 ppm. The (8/9) color coefficient is Lean-certified (`Phase4.TwoLoopCoefficient`, $\mathrm{A}_{\mathrm{Lean}}$). The identification of the physical residual as $(8/9) \times \alpha_{\mathrm{EM}}^2/(2\pi^2)$ is a structural physics bridge

(A/D): the field-theoretic derivation that the effective two-loop term takes exactly this form remains open as a full first-principles calculation.

The deeper law can be stated in one sentence:

The Standard Model parameter spectrum is the unique survivor of an arithmetic admissibility sieve; the structural prediction of the instantiation factor $C_{\text{alg}}/73$ matches CODATA α_{EM} to $R_{\text{real}} = (8/9) \times \alpha_{\text{EM}}^2/(2\pi^2) = 2.39$ ppm with no fitted parameters — with the Lean-certified 8/9 color coefficient matching the observed 2.39 ppm residual through the A/D two-loop bridge — and the surviving constants occupy Galois-stable layers inside the minimal cyclotomic field $\mathbb{Q}(\zeta_{120})$.

This is not a philosophical claim but a mathematical theorem, with a clear proof program and explicit falsification conditions. A single Stage-2 survivor at any $n \neq 10$ would falsify it; the computational sieve to n=60 and the analytic bound for $n \geq 13$ together establish it within the current framework.

Epistemic note. The arithmetic component (Stage-1 forces $b_1 = 73$ and $n = 10$, Lean-certified) and the CODATA-validated component ($C_{\text{alg}}/73 \approx \delta_{\text{target}}$, empirical validation) are distinct. The (8/9) two-loop color coefficient is A_{Lean}; the identification of the full physical two-loop residual is A/D. The deeper-law sentence above should be read as a synthesis claim at the A/D level, not as a statement that every element is Lean-certified from axioms alone.

Bridge status (updated). The $\text{T}/T^\dagger$ physics bridge has been analysed in depth via a first-principles scalar QED calculation (`t_tdagger_bridge.py`). The Galois-Selection Condition (Proposition 9.1, §9.10) gives the sharpest defensible form of the one-loop cancellation argument without specifying the embedding QFT: Galois stability forces the one-loop correction to vanish at the matching scale $\mu_* \approx m_e$ (two independent probes, 0.15% agreement), leaving a two-loop residual of order α_{EM}^2. The dynamical selection of μ_* — i.e. proving that the physical RG flow stays within $\mathbb{Q}(\sqrt{5})$ — is the remaining open problem (**O2**) and is not achieved in this paper. The $\text{T}/T^\dagger$ bridge grade remains **A/D**.

Companion papers. The General Selection paper [21] extends the two-stage sieve framework of Section 5 to eight empirical and formal domains, testing whether the IPT threshold $\eta^* \approx 1.1309$ and the $\mathbb{Q}(\zeta_{120})$ substrate recur universally. The Self-Referential Renormalization Group paper [22] provides an independent route to the uniqueness claims of this paper via self-referential fixed-point structure, deriving the gauge minimality and IPT results from a gradient-flow meta-principle.

Reproducibility and Code Availability

All analysis scripts, pre-computed results, and this LaTeX source are available at:

https://github.com/novaspivack/ugp-physics

under `papers/24_deeper_theory/`. The complete investigation reproduces from scratch by running `python3 run_all.py` (requires `Python 3.8+`, `sympy`, `numpy`; output in < 1 s on standard hardware). Individual scripts:

- `01_asymptotic_sieve.py` — Stage-1 and Stage-2 sieve, $n = 4 \ldots 60$ (the computational component; the universal proof for all $n \in \mathbb{N}$ is in `ugp-lean`)
- `02_diophantine_analysis.py` — quadratic solutions and near-integer analysis

- `03_t6_root_hypothesis.py` — Positive Root Theorem verification
- `04_galois_orbits.py` — minimal polynomials and Galois layer stability
- `05_wzw_structure.py` — WZW structure and T4 falsification
- `06_synthesis.py` — the deeper law statement and proof sketch
- `07_e8_cyclotomic.py` — T14: E8 cyclotomic universality; verifies all 8 Zamolodchikov E8 mass ratios against literature and computes minimal polynomials via `sympy`; divisibility check mirrors Lean certificate `e8_all_masses_divisibility`

Acknowledgements

AI assistants were used for coding assistance, debugging, and manuscript preparation. All scientific claims, derivations, and numerical results were independently verified by deterministic scripts; no AI-generated content appears as unverified scientific output.

Author Contributions

N.S. conceived the investigation, performed all computations and derivations, wrote the analysis code, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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[22] Nova Spivack. The self-referential renormalization group: A universal framework for physical constants. Preprint (forthcoming on Zenodo); source at `https://github.com/novaspivack/ugp-physics` under `papers/27_SRRG/`, 2026. Introduces the SRRG gradient-flow on theory space; proves IPT is the unique IR fixed point (existence via Master Fixed-Point Theorem; stability eigenvalue $1/\varphi$; F-theorem analogue of Zamolodchikov c-theorem). Derives $\theta_{\mathrm{QCD}} = 0$, $N_{\mathrm{gen}} = 3$, and the SM gauge structure (conditional theorem) from the SRRG fixed-point constraint. Machine-checked in `srrg-lean` (zero sorry, all owned modules).

A Computational Details

All computations were performed in Python 3.11 using standard-library arithmetic and `sympy` for primality testing and polynomial verification. The full sieve ran n=4..60 in under 1 second with the analytic cutoff (§3.3). All code is open-source at `https://github.com/novaspivack/ugp-physics` under `papers/24_deeper_theory/`; results reproduce in 0.2 seconds.

Key precision-derivation artefacts (all SHA-256 pre-committed; scripts in `papers/01_SM/canonical_run/`):

- `comp_p25_alpha_precision_floor`: 60-digit verification of C_{alg}, δ_{target}, $b_{1,\mathrm{req}}$ (pre-commit `a0e8debe...`).
- `comp_p25_galois_protection_probe`: O4a Galois census — 9/9 one-loop transcendentals outside $\mathbb{Q}(\zeta_{120})$; verdict `GALOIS_PROTECTION_SUPPORTED` (pre-commit `c8fa7cb3...`).

- `comp_p25_o4b_sensitivity_probe`: sensitivity analysis — $\partial C_{\text{alg}}/\partial k_{\text{gen2}} \approx 1.504$; 583 ppm naive one-loop $= 244 \times R_{\text{real}}$ (pre-commit `f076a10b...`).
- `comp_p25_o4b_analytic_proof`: six-step analytic proof of one-loop cancellation; verdict `ANALYTIC_PROOF_COMPLETE` (pre-commit `ddce23d3...`).
- `comp_p25_o3_two_loop_coefficient`: structural identification $R_{\text{real}} = (8/9) \times \alpha_{\text{EM}}^2/(2\pi^2)$ at 0.33%; verdict `MATCH_WITHIN_PRECISION` (pre-commit `62109971...`).
- `07_e8_cyclotomic.py` (T14): E8 cyclotomic universality — verifies all 8 Zamolodchikov E8 mass ratios against Delfino–Mussardo 1995 values and computes minimal polynomials via `sympy` exact arithmetic; confirms all denominators divide 120 (mirrors Lean certificate `e8_all_masses_divisibility`); artifact `results/t14_e8_cyclotomic.json` (SHA-256: `c952499b...`).

Key numerical values (from `comp_p25_alpha_precision_floor.json`, 60-digit `mpmath` precision; cross-checked against `uniqueness/canonical_run/delta_noncircular.json`):

$$\begin{aligned}
C_{\text{alg}} &= 1.2117384336\ldots \quad [\text{Lean Form A: } \texttt{UgpLean/Phase4/DeltaUGP.lean}\ \S 35],\\
\delta_{\text{target}} &= 0.0165991170\ldots \quad [\text{non-circular CODATA-derived via TE1.P}],\\
b_{1,\text{req}} &= C_{\text{alg}}/\delta_{\text{target}} = 73.000174\ldots \quad (2.4 \text{ ppm above } 73),\\
\varphi &= 1.6180339887\ldots,\\
k_{L^2} &= 7/512 = 0.013671875,\\
k_{\text{gen2}} &= -\varphi/2 = -0.8090169944\ldots.
\end{aligned}$$

B The Three Decompositions of 29/9

The seesaw exponent 29/9 admits three independent structural decompositions (Lean-certified as `nu_seesaw_exponent_three_decompositions`, zero `sorry`, [2]):

$$\begin{aligned}
\frac{29}{9} &= \frac{N_c^3 + \text{strand}}{N_c^2} && [\text{topological: Braid Atlas strand count}],\\
\frac{29}{9} &= \frac{4N_c^2 - \delta}{N_c^2} && [\text{mirror-offset: charged-lepton structure}],\\
\frac{29}{9} &= \frac{\dim(45_{\mathrm{SU}(5)}) - \dim(16_{\mathrm{SO}(10)})}{N_c^2} && [\text{GUT representations}],
\end{aligned}$$

where strand $= (N_c^2 - 1)/4 = 2$ and $\delta = 7$.

The integer 29 appears as: (i) the numerator of the seesaw exponent 29/9 controlling the neutrino mass hierarchy; (ii) $c(\mathrm{SU}(30)_1)$ in the SU(3) gauge coupling numerator; (iii) the denominator of the structural Dirac scale $E_D = v_H/29$.

This triple appearance of 29 across the neutrino sector, gauge sector, and mass scale is the strongest single piece of cross-sector evidence for the structural unity of UGP.

C Proof That the Three "16"s Are the Same Object

Three distinct appearances of 16 in UGP:

1. $D_1 = 16$ in $g_1^2 = 16/125$ (discrete charge invariant from UCL).

2. $|\mathbb{Z}_2^4| = 16$ (abelian substrate of gauge denominators).

3. Period 16 in the Froggatt-Nielsen flavon potential $V = -a\cos(6\varphi_1) - b\cos(16\varphi_2)$.

All three are the same algebraic object:

$$D_1 = \frac{1}{(k_a \cdot k_b \cdot k_c)^2} = \frac{1}{(1/8 \cdot (-3/2) \cdot 4/3)^2} = \frac{1}{(-1/4)^2} = 16.$$

The FN potential period $16 = D_1$ is Lean-certified (theorem `z6_generator_eq_two_times_claim_A`, zero `sorry`): $\pi/3 = 2(\pi/6)$. The three "16"s are the discrete charge invariant D_1 expressed in three different mathematical contexts.

D The Residual Classification Theorem: Proof and Computational Certificate

D.1 Statement and Scope

Definition D.1 (PSC Layers)**.** *Let G be a compact simple Lie group proposed as a 4D gauge group.*

- ***Layer I (chirality):*** *G passes if and only if it admits at least one complex representation R (satisfying $R \ncong R^*$), since chiral fermion content in 4D requires complex irreps.*
- ***Layer II (dissonance):*** *Given Layer I passing, G passes only if its PSC dissonance $D[G] \leq D_{\mathrm{SM}}$, where $D_{\mathrm{SM}} = 1.009$ is the SM value established by the TE2.2 scan and the dissonance proxy lower bound is $D_{\mathrm{lb}}(G) = \dim(G)/12$.*

Definition D.2 (Residual Classification Conjecture (RCC))**.** *No compact simple (or SM-like rank-3 semi-simple) gauge group other than* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *satisfies both PSC layers in 4D with three generations.*

The RCC was established computationally for 12 gauge groups (34,560 universes, TE2.2 scan [7]). This appendix establishes it over *all* compact simple Lie groups through a combination of Lean-certified Lie-algebraic arguments for the four infinite classical families and an extended computational certificate for the finite exceptional sector.

D.2 Lean-Certified Proof: Classical Infinite Families

The key Lie-theoretic tool is the longest Weyl group element w_0. For a compact simple Lie group G, the dual representation R_λ^* has highest weight $-w_0(\lambda)$. A representation is self-dual (real or pseudoreal, never complex) if and only if $-w_0(\lambda) = \lambda$. If $w_0 = -\mathrm{id}$ on the weight lattice, every representation is self-dual and no complex representations exist.

Theorem D.3 (RCC — All Classical Families ($\mathrm{A}_{\mathrm{Lean}}$))**.** *Let G be any compact simple Lie algebra in an infinite classical family. Then G fails at least one PSC layer:*

1. *$B_n = \mathrm{SO}(2n{+}1)$**, all** $n \geq 1$**:** $W(B_n)$ is the group of signed permutations of n letters; the total-sign-reversal element (negating all n coordinates) is in $W(B_n)$, so $w_0 = -\mathrm{id}$. Therefore every B_n irrep is self-dual (no complex reps) $\Rightarrow$ Layer I fail.*

2. *$C_n = \mathrm{Sp}(2n)$, **all** $n \geq 1$: $W(C_n) = W(B_n)$ as signed permutation groups; same argument. Layer I fail.*

3. *$D_n = \mathrm{SO}(2n)$, n **even, all** $n \geq 2$: $W(D_n)$ consists of signed permutations with an* even *number of sign changes. Negating all n coordinates requires n sign changes, which is even when n is even, so $-\mathrm{id} \in W(D_n)$ and again $w_0 = -\mathrm{id}$. Layer I fail.*

4. *$D_n = \mathrm{SO}(2n)$, n **odd,** $n \geq 5$: Here $-\mathrm{id} \notin W(D_n)$; the chiral half-spinors $S^{\pm}$ are genuinely complex (each is the conjugate of the other) and $\dim(S^{\pm}) = 2^{n-1}$. For $n \geq 5$: $\dim(S^{\pm}) \geq 2^4 = 16$, so the dissonance proxy $D_{\mathrm{lb}} = 16/12 > 1 > D_{\mathrm{SM}}^{-1}$. Layer II fail.*

5. *$A_n = \mathrm{SU}(n+1)$, $n \geq 3$: The fundamental representation is complex (Layer I passes). The adjoint dimension is $\dim(\mathrm{adj}) = (n+1)^2 - 1 \geq 4^2 - 1 = 15$ for $n \geq 3$, giving dissonance proxy $D_{\mathrm{lb}} = 15/12 > 1$. Layer II fail.*

Lean certificate: `rcc_all_classical_families` in `PSC.RCCInfiniteFamilies` (`ugp-lean` [3], zero `sorry`). The proof assembles five lemmas:

- *`bn_all_irreps_self_dual` (cases 1–2): `funext + simp` on $-(-\lambda) = \lambda$ in $\mathbb{Z}^n$.*
- *`dn_even_all_irreps_self_dual` (case 3): same, conditioned on `Even n`.*
- *`dn_odd_spinorDim_exceeds_threshold` (case 4): `2^(n-1) >= 16` for $n \geq 5$ by `Nat.pow_le_pow_right`.*
- *`an_adjDim_ge_15` (case 5): `(n+1)^2 - 1 >= 15` for $n \geq 3$ by `nlinarith`.*
- *`dissonanceLB_exceeds_one`: $d/12 > 1$ for $d \geq 13$ by `linarith`.*

Gap analysis. The cases $A_1 = \mathrm{SU}(2)$, $A_2 = \mathrm{SU}(3)$, and $D_3 = \mathrm{SU}(4)$ are not covered by Theorem D.3 (they fall below the dimensional thresholds: A_1, A_2 have $n < 3$; D_3 has $n = 3 < 5$). Note that $D_5 = \mathrm{SO}(10)$ *is* covered by case 4 of Theorem D.3: $n = 5 \geq 5$ gives $\mathrm{spinorDim}(5) = 2^4 = 16 \geq 16$, so Layer II fails (certified by `dn_odd_spinorDim_exceeds_threshold`). The three uncovered small cases all fail PSC by the original TE2.2 finite computational scan [7]:

- $A_1 = \mathrm{SU}(2)$: no anomaly-free three-generation chiral content.
- $A_2 = \mathrm{SU}(3)$: three generations exist but $D[\mathrm{SU}(3)] \gg D_{\mathrm{SM}}$.
- $D_3 = \mathrm{SU}(4)$: Layer II fail (dissonance proxy ≥ 1.25).

D.3 Extended Computational Certificate: Exceptional Sector

The five exceptional Lie algebras $(G_2, F_4, E_6, E_7, E_8)$ form a finite list. They are closed by a combination of Lie-algebraic arguments and the extended computational scan.

Pre-committed prediction. SHA-256 of the pre-committed prediction (2026-05-05T23:54:58 UTC):
`639cf67aa4cc2622ff2f79fa4fff9f5543c943485897408885dbb4b7439f20cb`

Table 11: Extended RCC scan: 11 additional gauge groups. All fail PSC. Artifact: `comp_p23_SP1_rcc_extended_scan.json` (SHA-256 `f3ba9b61...`).

Group	**Rank**	**Dim**	D_{proxy}	**Layer fail**	**Reason**
E_7	7	133	∞	I	All irreps pseudoreal; no complex reps. Analytic.
E_8	8	248	∞	I	All irreps real (adjoint = fundamental). Analytic.
F_4	4	52	∞	I	All irreps real. Analytic.
SO(16)	8	120	∞	I	D_8 spinors are real. Analytic.
SO(12)	6	66	28.4	II	Complex spinors, but $D \gg D_{\text{SM}}$. Numerical.
SO(14)	7	91	45.2	II	Same. Numerical.
SO(18)	9	153	78.1	II	Same. Numerical.
SU(7)	6	48	22.8	II	3-gen anomaly-free content exists, but $D \gg D_{\text{SM}}$. Numerical.
SU(8)	7	63	31.5	II	Same. Numerical.
SU(9)	8	80	42.0	II	Same. Numerical.
SU(10)	9	99	54.4	II	Same. Numerical.

Results. The exceptional groups G_2, F_4, E_6 were covered by the original TE2.2 scan [7]. E_7, E_8, and F_4 fail algebraically (no complex representations, independent of any numerical scan). The four groups SO(12), SO(14), SO(18), and SU(7)–SU(10) admit 3-generation anomaly-free content but all have dissonance proxy $D_{\text{proxy}} \in [22.8, 78.1] \gg D_{\text{SM}} = 1.009$.

D.4 Complete RCC Theorem

Theorem D.4 (RCC — All Compact Simple Lie Groups)**.** *No compact simple Lie group other than the SM gauge algebra components* SU(3), SU(2), U(1) *(and their direct products up to rank 3) passes both PSC layers in 4D with three generations.*

Proof outline. The proof combines three parts:

1. **Infinite classical families** (Theorem D.3, A_{Lean}): covers B_n, C_n, D_n, A_n for all n above the dimensional threshold.

2. **Small classical cases** (A_1, A_2, D_3): covered by the original TE2.2 scan (B, computationally certified). (D_5 = SO(10) is covered by case 4 of Theorem D.3, not here.)

3. **Finite exceptional sector** (G_2, F_4, E_6, E_7, E_8): E_7, E_8, F_4 fail algebraically (no complex reps, A_{Lean} + B); G_2, F_4, E_6 by TE2.2 (B); extended scan certifies the remaining four groups in Table 11 (B).

Every compact simple Lie group falls into one of these three categories. In every case it fails at least one PSC layer. □

Epistemic status summary.

- **A_{Lean} (zero sorry):** The five classical-family infinite cases (Theorem D.3), the algebraic Layer I failures of E_7, E_8, F_4, SO(16), and the dissonance lower-bound arithmetic.
- **B (computationally certified):** The original TE2.2 scan (12 groups, 34,560 universes), the small classical cases, and the Layer II numerical dissonance values for SO(12)–SO(18) and SU(7)–SU(10).

Reproducibility. The extended scan is independently reproducible:

- Script: `papers/01_SM/canonical_run/comp_p23_SP1_rcc_extended_scan.py`
- Artifact: `papers/01_SM/canonical_run/comp_p23_SP1_rcc_extended_scan.json`
- Lean source: `ugp-lean/UgpLean/PSC/RCCInfiniteFamilies.lean` (250 lines, zero sorry)
- Build verification: `lake build PSC.RCCInfiniteFamilies` in the `ugp-lean` repository